International Association of Oil & Gas Producers

REPORT
2021sh

JULY
2022

DATA SERIES

Safety performance indicators – 2021 data – High potential event reports

Acknowledgements

IOGP acknowledges the participation of the companies that have submitted safety performance indicators.

This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

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DATA SERIES

Safety performance indicators – 2021 data – High potential event reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	July 2022	First release

Contents

AFRICA ONSHORE

DATE: **Nov 18 2021**
COUNTRY: **Algeria**
FUNCTION: None
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

At around 08:00 hrs., a subcontractor driver overturned a truck while navigating a roundabout. Two IBCs containing brine were spilled.

The volume of brine discharged is estimated to be approximately 700l.

Immediate actions:

- Return the IBCs to its vertical position and clear the road.
- Rinsing with road water.

WHAT WENT WRONG:

Management: inadequate supervision:

- The supervisor at the site had not inspected the loading and securing before the journey.

Equipment: inadequate transportation or handling.

- The strap was in poor shape and the load was not secured properly, if at all.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Following the course of the investigation, interviews, and site visit, the following emerge:

- The strap used was in poor condition (non-compliant).
- The trailer was in a degraded state (The problems related to the truck had already been raised with the subcontractor on October 31 by the site supervisor, but no firm decision was taken to stop it).
- Poor securing of IBCs

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: **Feb 20 2021**
COUNTRY: **Algeria**
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

During the mechanical de-isolation of a vessel, a 24 inch blind, weighing approximately 200 Kg, was being lifted from the top of the vessel to the ground.

The route of the lift was not adequately cleared and needed to be paused by HSE to allow personnel to be removed from under the load path.

AFRICA ONSHORE

When the load reached the ground it was found that there was no lifting eye on the blind. A scaffold clip had been attached to a protruding part of the blind and the sling was only prevented from slipping of the blind by the scaffold clip.

This is in breach of normal lifting practices, using a scaffold clip as a lift securing method and could have resulted in the 200 kg blind dropping over 20 metres.

Immediate action taken- discussed with the work party involved and removed the scaffold clip from the blind.

WHAT WENT WRONG:

- Management: inadequate supervision.
- Inadequate Design: spacer designed without lifting lugs.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure suitable lifting points are available before allowing on site.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Jan 1 2021
COUNTRY: Algeria
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

At 13:50 hrs., during a well kill operation under a PTW with toolbox talk attached to it, a piece of grating was removed for a quick access to the well cellar in order to exchange hand tools despite the fact that the well cellar was designed with specific access with a ladder.

The supervisor was installing a pressure gauge fully equipped with PPEs on the well head when he lost his balance and fell into the cellar on the sand that covers the bottom.

The victim was evacuated immediately to the clinic for medical check and first aid treatment. According to the doctor the victim is in good condition with a small scratch on the nose.

WHAT WENT WRONG:

- Hazard identification.
- Lack of appropriate management of change.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Do not change the way you are performing a task without a proper risk re-assessment.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

AFRICA ONSHORE

DATE: **Dec 8 2021**
COUNTRY: **Algeria**
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

At 15:40pm, the construction team started the repair job on the production line at PV manifold in order to cut the corroded part of the pipe. As soon as they started to open the clamp to get it removed (previously put in place), water started to come out and after just a few seconds a huge amount of gas escaped.

Stopped all activities immediately and started to evacuate the area. At 15:50, the intervention agent present on the location started to spray water on the gas leaking area to contain the gas and minimize the spread of the leaking gas. At the same time, intervention team has been informed to get to the affected area immediately.

At 16:00 alarms have been activated and order from Exploitation Manager to for total Shut Down has been given. All personnel have been evacuated and mustered.

Intervention team continued spraying water and monitoring the gas amount. The area had been barricaded and access to the CPF had been banded for everybody except HSE until full depressurization of the line.

At around 17:35, the area was clear and permit has been given to Technical and Exploitation personnel to get to the area for securing the line and to check visually positive isolations and gathering preliminary information.

At 19:35, order was given to re-start the CPF.

No personal injuries or equipment damage occurred.

WHAT WENT WRONG:

- Inadequate permit to work mitigation measures identification.
- Inadequate/defective guards or protective barriers.
- Inadequate design/specification/management of change (pipeline was flushed but water dilution contained gas, due to the modification to introduce water in the pipeline).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Management of Change is required in any modification impacting the normal operating conditions.
2. Competency assurance system is required to ensure proper risk identification and management.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

AFRICA ONSHORE

DATE: **May 18 2021**
COUNTRY: **Algeria**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Safe mechanical lifting

NARRATIVE:

While conducting a lifting operation to pick up an extra drill pipe, the drill pipe joint disconnected from the lifting cup, hit the cat walk, stopped at the end of cat walk stopper, and finally fell on the ground.

WHAT WENT WRONG:

Inadequate connecting procedure and tool used.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Before rigging up any tool in the rig floor (anchored, pinned, lifted, etc), the third party Supervisor must verify conformity and communicate it to the rig floor crew.

Weekly rig inspection must include anchor and liftings points.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

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DATE: **Mar 23 2021**
COUNTRY: **Algeria**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Safe mechanical lifting

NARRATIVE:

During the slickline operation, the Contractor operator anchored one sheave in the wrong pad eye. When they started to apply tension on the cable to pull free from the top of liner, a part of the wooden plate was lifted out from its initial position and travelled down the slickline cable along with the sheave making a complete stop in front of the slickline unit. No injuries or property damage.

WHAT WENT WRONG:

- Improper Communication.
- Improper supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Before rigging up any tool in the rig floor, 3rd party supervisor must verify conformity and communicate it to the rig floor crew.
- Weekly rig inspection must include anchored and lifting points (visualize with different colours).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

AFRICA ONSHORE

DATE: **Sep 21 2021**
COUNTRY: **Algeria**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

In the main camp (between Admin building and clinic), an electric pole suddenly fell down due to corrosion on it base.

WHAT WENT WRONG:

Inadequate maintenance program. Inadequate hazard identification.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Preventive maintenance is also key for structures in the main camp, taking into account extreme weather conditions.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: **Aug 21 2021**
COUNTRY: **Angola**
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

While performing cleaning duties inside a compressed air skid container, the operator raised the air hose (120 PSI) when, suddenly, the valve burst, causing the hose to disconnect, twist and make the safety latch come off. The hose swung, reaching the operator and hitting him on the abdomen.

WHAT WENT WRONG:

- Broken valve body thread.
- Inadequate equipment integrity.
- Ring fatigue.
- The safety cable was not linked to a fixed point that would prevent it from moving.
- No record of valve inspection or replacement since 2008.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Verify the materials´ and fastenings devices' condition.(those that are frequently used).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

AFRICA ONSHORE

DATE: Dec 4 2021
COUNTRY: Congo
FUNCTION: Production
CAUSE: Confined space
ACTIVITY: Maintenance, inspection, testing
RULE: Confined space

NARRATIVE:

Two workers equipped with cartridge filter masks doing painting works in a void space failed to respond to a routine communication check. The first worker evacuated on his own and the second one was unable to respond correctly to instructions to vacate the space. He became further unwell and unable to evacuate on his own. A rescue team eventually established a fresh air supply to the IP. He partially recovered and was able to vacate the space assisted by the rescue team. Due to the extremely challenging location in which he was working the rescue effort took three and a half hours.

WHAT WENT WRONG:

Insufficiently ventilated void. The void space was saturated with toxic vapours.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Raise awareness of the risks associated with work in confined spaces.
2. Improve the process of carrying out AROs during interventions in confined spaces .
3. Procedure should be adapted (number of painted compartments).

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: Jul 20 2021
COUNTRY: Gabon
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

At around 06:38 hrs., floor men broke out a connection in the BHA, removed the slips and moved from the red zone to a safe place. While pulling out of hole BHA, at 19m high the cylindrical pin which retains the upper part of the hydraulic piston tilt of the TDS of about 700g weight came out from its axis and felt to the rig floor (on the red zone), rebounded within a metre from a floor man standing at 4 metres away from the rotary table, rebounded again and got lodged in the structure of the ODS winch. After the pin came out from its housing, the tilt piston became loose from the top of the TDS, slipped and finished retained by a hydraulic hose. The cotter pin retaining the pin was crushed but still in place.

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WHAT WENT WRONG:

Immediate causes:

Drilling operation produces vibrations.

The cylindrical pin was no longer well positioned in its housing.

Pre- conditions:

No visual inspection performed prior to start operation (not required as handling BHA is a routine operation).

Failure to identify the cylindrical pin not in its housing when broke out BHA and handling slips.

Underlying causes:

Vibrations are being produced during daily drilling operations. Monthly mast inspection is done but is not sufficient and not sufficiently documented.

Lack of vigilance by the crew, when TDS is at the rig floor level, crew should identify or have a close look at any parts of the TDS that could have moved.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Weekly inspection will be implemented and all documents related kept in a binder.
- Hazard recognition awareness to be implemented.
- Reinforce training on drop object.
- Develop and Share learning from incident.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **May 20 2021**
COUNTRY: **Gabon**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

On May 20, 2021, at approximately 05:00 am, the driver truck was traveling with his helper in a Truck.

Whilst driving the helper saw buffalos crossing the road and informed the driver by touching him; driver tried to swerve to avoid them but lost control of the vehicle, resulting in the vehicle leaving the road and overturning off road.

No Injuries.

Alcohol Test – Negative.

Equipment damaged.

WHAT WENT WRONG:

Root Cause:

Logistics crew had poor sleep cycle - Fatigue.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Reduce workload for logistic night crew and maximum operations have to be done during day and better planning for night shift operations.
- Have more personnel capable to drive the trucks in workforce.
- Ensure Implementation of Fatigue Management awareness every 3 months for crew.
- Driver must do Defensive Driving Test prior to be allowed to drive again on location.
- No Zig-Zag Driving allowed on location and not try to swerve when see animals at the middle of the road.
- Review the content of Contractor DDC module in line with Company requirements.
- Fatigue Management Refresh to be done for dedicated drivers at the beginning of their rotation on site.
- Sharing Information, Alert/Learnings of all Incidents occurring.
- Logistics Team Lead to arrange on a daily basis with the crew some adequate rest times following long and short driving duration.
- Reduce the delay for demobilizing unnecessary personnel from site.
- Review catering time schedule to ensure adequate rest of Logistics teams.
- Provide rest room for Logistics crew during standby period.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

DATE: Jun 3 2021
COUNTRY: Gabon
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

At the central station at approximately 3:12 am, a fire was observed by operator on the P-2506 and P-2508 oil export pumps after receiving fire alarm from the DCS screen.

The alert was given by phone and the station was shut down at 3.13 am. The 3 operators quickly controlled the fire using extinguishers.

Following authorization from management and site checks, the station was re-started smoothly on the same day at approximately 11:00am.

No one was injured. One pump was damaged as a result of the fire.

WHAT WENT WRONG:

The most probable root cause is seen as follow:

- P2506 bearing cover (loosening of the bolts by vibration, absence of pinning, or both not being present at the beginning), migrate downwards shaft, generating a metal-to-metal contact between the cover and the rotating pump shaft. This friction causing a hot spot defined as the source of activation of the fire.
- The abrasion of the cover by the presence of sand which was driven into the crude oil generated a leak which, on contact with the hot spot, ignited and, by projection effect on the P2508, ignited the pump and the stagnant oil crude present at the skid bund/and from leakage point from instrument tie-in on pump.

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- The investigation outlines that the damage of P2508 is less than the damage of P2506 due to the difference in the skid of each pump.
- Also the secondary effects of the CS staff transition in April/May led to inadequate monitoring and cleaning/ close monitoring measures. The satisfactory level of housekeeping was seen below previous practices.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Provide installation of a pin on each pump.
- Confirm the presence of the pin on each pump and provide for modification if not exist.
- Check and confirm the current design versus initial design.
- Replicate the design of the P-2508 skid (protective plate between the lower part of the bundwall and the pump) on the other pumps to avoid escalation of fire in the event of a fire.
- Install bearing temperature on each pumps (oil export+booster pumps).
- Re-design the collector system to avoid liquid spread outside (drip pan should collect all the liquid from quench seal).
- Adjust the bi-weekly PM to control and verify tightness of bolts.
- Adjust the frequency of the vibration controls.
- Extinguisher firefighting training refresh to be delivered to all operation and maintenance personnel.
- Refresh the emergency initiation to all central station personnel.
- Complete the fire system export trip project to trip all the export skid (actually on alarm only).
- Implement a temperature detector on each pump bearing (oil export and booster pumps).
- Re-evaluate the housekeeping scope and reinforce staffing if required.
- Internalization of Vibration measurement and control: provide material and adequate training to the technicians.
- Implementation of a daily temperature control of the pump coupling (as first line maintenance).
- Implement a sand management strategy at central station oil export and booster pumps to avoid filter clogging.
- Organize a routine temperature check with handheld thermometer until fix sensors are installed.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: Feb 4 2021
COUNTRY: Nigeria
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
RULE: Other issue

NARRATIVE:

At about 09:30 hrs., while inflating a tyre of a front wheel loader, a Shop Maintenance Contract personnel suffered a fracture injury to both the ulnar and radius bones of his right forearm when the outer ring flange pulled out during inflation, hitting his arm.

The Injured Person (IP) was immediately taken to the clinic where he was stabilized and later moved to Company Hospital for further medical care.

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WHAT WENT WRONG:

- Design did not consider human factors.
- Inadequate Maintenance.
- Procedures not available.
- Inadequate work oversight or enforcement of work standards.
- Training exists but inadequate.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: Feb 9 2021
COUNTRY: Nigeria
FUNCTION: Unspecified
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

The incident occurred while a 10T overhead crane was being used to relocate a 31.5 ft, 530kg, 8" export suction hose from the aisle to an open area within the warehouse. While lifting the suction hose, one end of the hose got stuck on the floor, while the other end got stuck in an adjourning mooring hose rack. The IP put his hands around the suction hose attempting to free the flange end that was stuck on the floor by pulling the hose towards himself while the Crane Operator was trying to free the hose with the overhead crane. The suction hose recoiled towards the IP, making him to fall backward, and the flange end of the hose fell on his right foot.

WHAT WENT WRONG:

- Wilful deviation.
- Management expectations inadequately documented, communicated or enforced.
- No Risk Assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate communication

AFRICA ONSHORE

DATE: Jun 5 2021
COUNTRY: Tunisia
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Work authorization

NARRATIVE:

Whilst pulling out of hole (POOH) the sucker rod (SR) string, the Chief Mechanic found the chain of the Twin Stop safety system disengaged and re-installed it, with no further tests and checks.

During lay down of one SR single at around 4 metres from the rig floor the twin stop activated suddenly and stopped the traveling block (TB). The brake of the drawwork was freed by adjusting the equalizing bar. Then, continued the POOH. When stopped to break out the SR joint at around 8 metres, the TB started to drop. Assistant Driller pushed hard on the brake but did not stop the fall of the TB. Tool Pusher activated the emergency braking system (crown-o-matic), which also failed to brake as its support came off due to the blockage of the drawwork braking shaft (caused by the bending of one brake transmission rod occurred earlier).

WHAT WENT WRONG:

- Failure of the drawworks Breaking system due to manipulation by chief mechanic.
- Chief mechanic did not inform WO supervisor - break down of communication.
- No JSA or PTW done - lack of operational discipline.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Improve compliance to rig auditing standard regarding SCE inspection certification based on OEM requirements, safety regulations and equipment functionality.
- Design review of the twin limit system functionality (crown and floor saver) and consider upgrade with the aim to ensure the system is fail safe.
- Improve Safety culture implementation and communications with clear actions.
- Review possibility to install CCTV surveillance of critical rig area.
- Install Real time data recording for WO campaign before re-starting.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate communication

AFRICA ONSHORE

DATE: Jun 28 2021
COUNTRY: Tunisia
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

The field operator (eye-witness) noticed a white cloud being released from the X-tree. The field operator managed quickly to inform via VHF radio the Control Room Operator (CCR) for shutting-in the well. The later activated, few seconds later, a Level 4 emergency shutdown that closed the SSSV, the actuated master valve and the actuated wing valve which reduced the flowrate.

WHAT WENT WRONG:

- Substandard situations are accepted, deviation tolerated.
- Operating outside the envelope for sustaining the production targets.
- Poor of organizational management of change and competency management combined by the pandemic "abnormal".
- Unclear accountability, reporting lines, job specifications and command chain.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Encourage the reporting culture and the Stop work policy! and further promote Life-Saving
- Rules (LSR)
- Enhance the HSSE Leadership across all levels.
- Implement operational management of change impacting the CPF original design.
- Develop the missing operating procedures for exceptional tasks.
- Identify other potential deviations vs standards for the current operations.
- Relaunch the CPF Competency Management System including the remaining EPCC, vendor trainings and internal trainings.
- Clarify EDP and Asset teams' responsibilities and accountabilities via clear job descriptions, reporting lines and chain of command.
- Reinforce the back office support (e.g., Well integrity and maintenance).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation intentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

AFRICA OFFSHORE

DATE: **Nov 4 2021**
COUNTRY: **Nigeria**
FUNCTION: Production
CAUSE: Explosion
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

A gas leak occurred following the rupture of a manual valve located on a fuel gas line supplying a Turbine Generator whilst operating at around 55 bars. The rupture was seen to be due to the fracture of a number of bolts located on the failed valve. All personnel mustered. Production shortfall and excess flaring associated to this event.

WHAT WENT WRONG:

Leak due to broken manual valve. Bolt material not suitable.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Assembly should have been aligned to avoid piping damaged in operation.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **Feb 7 2021**
COUNTRY: **Nigeria**
FUNCTION: Production
CAUSE: Water related, drowning
ACTIVITY: Transport - Water, incl. marine activity
RULE: Line of fire

NARRATIVE:

At about 0650hrs, a crew member on an Export Tug sustained a hand injury when his right-hand wrist got trapped between a handrail and vessel body while unmooring the vessel from the stand-by buoy near the BOP. The operation was suspended, and the incident immediately reported. The injured person was evacuated to the clinic for medical attention.

WHAT WENT WRONG:

- Design did not anticipate the conditions.
- Management of Change not utilized.
- HES Procedures or Safe Work Practices not utilized.
- Operations Procedures not available.
- No Risk Assessment.
- Inadequate work direction or unclear expectations.

AFRICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

ASIA/AUSTRALASIA ONSHORE

No high potential incidents reported

ASIA/AUSTRALASIA OFFSHORE

No high potential incidents reported

EUROPE ONSHORE

DATE: Feb 19 2021
COUNTRY: Germany
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Other issue

NARRATIVE:

A farmer reported an anomaly at the pipeline to the control centre.

The damaged area was visited and identified.

Pressure release through gas treatment facility.

Shut off the pipeline section in the field station.

Information passed on to regional mining authorities and regional public.

After submission of the results by the environmental analysis experts, a distinctly lower gas discharge could be established.

WHAT WENT WRONG:

Equipment: inadequate maintenance or inspection in accordance with an approved maintenance or inspection regime.

Corrosion.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Feb 20 2021
COUNTRY: Norway
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

During scaffolding construction in ballast tank 2 starboard, a 3 metre long scaffolding plank detached from the rope which was used to lower scaffolding material down through manholes. The scaffolding plank hit the left foot of the IP, partly on the shoes that had a protective toe and midfoot. Height down to the repo was 6 metres. Estimated fall height approximately 3 metres with a weight of 11.2 kg. Energy estimated to 329 Joules. IP got a cut on the foot and was transported to the emergency room where 3 stitches were sewn. No other damage revealed. IP sent back to accommodation and ordered to rest for four days. Similar activities were stopped in other tanks in order to verify correct handling.

EUROPE ONSHORE

WHAT WENT WRONG:

Improper lifting, loading, and/or placement.

Scaffolding plank was attached using quick shackle to lock rope. When the plank was lowered over the edge of the manhole and had come some distance down, the plank slipped out of the loop.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Scaffolding plank detached from rope while being lowered into manhole and hit person on level below. The work was insufficiently planned which led to the plank not being secured with the appropriate safety hook, and the recipient had to stand in unfavourable position.

RECOMMENDATIONS:

- Purchase safety hooks that fit all scaffolding material.
- Implement Best practice for handling and securing scaffolding equipment at height.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: Jan 20 2021
COUNTRY: Norway
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

A scaffold pipe weighing 10 kg was lost during handling and fell 18 metres onto the quay side. Impact energy 1765 Joule.

The quay side was not barriered off, although it was marked with signs as "shielded". There were personnel working in the area, although not in the immediate vicinity of the impact point.

WHAT WENT WRONG:

- Improper lifting, loading and/or placement.
- Inadequate guards/signing/cordoning.

A scaffold operator lost control of the Scaffold spire while placing the spire on top of existing spire, which resulted in losing the Scaffold spire overboard and fell to the quay side below.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Scaffolding part was dropped while building a scaffolding due to poor technique and inexperienced scaffolder crew. This issue was not identified job was not sufficiently planned or risk assessed. SIMOPS was not identified which led to area below not being cordoned off.

RECOMMENDATIONS:

- Strengthen follow-up of subcontractors through pre-qualifications and audits.
- Increase use of SJA when inexperienced personnel is involved.
- Implement requirement for work permits for scaffolding work.
- Implement standardized guidelines for scaffolding.

EUROPE ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: **Mar 31 2021**
COUNTRY: **Norway**
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

During lifting and assembling protection hatch, crane operator lifted and placed the hatch without locking the hinge. During the lifting and placement of second hatch, the first hatch released from the hinge and fell to ground. This resulted in some cosmetic paint damage to the hatch.

During the lifting operation, there were no contractor employees present. According to the crane operator, there were employees from other companies present in the hall who were involved, and they gave instruction to crane operator to lift and assemble the hatch. According to the crane operator, lifting and assembly was done to continue with testing of other glass fibre hatches (which is in the middle). Lifting drawing with sling arrangements and sling length was with crane operator during the operation. Weight of Hatch approximately 8 tons. Height of fall approximately 3.5 metres. Under slightly changed circumstances there could have been a fatality.

WHAT WENT WRONG:

Failure to identify risk.

Improper lifting, loading, and/or placement.

Inadequate guards/signing/cordoning.

Requirement (procedure, process etc.) not followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

During lifting and assembling of protection hatch, the hatch fell to the ground. Several mistakes were made during the operation, caused by insufficient handover, ineffective communication regarding change of plans, SJA/TBT not conducted and refresher training not provided to crane operator.

RECOMMENDATIONS:

- Develop handover process and communication plan.
- Review and improve SJA process and requirements.
- Implement mandatory TBT for critical activities.
- Reinforce requirements for barriers during lifting operations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE ONSHORE

DATE: Mar 22 2021
COUNTRY: Romania
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

On 22nd of March 2021, at approximately 11:45 workover crew was pulling out of hole hydril tubing when wire rope ruptured and travelling block together with tubing string fell uncontrolled from a height of minimum 5 metres. As a result, the travelling block weighing 1.5 tonnes fell on the working platform and one employee was thrown on the ground, suffering fractures to one hand and one leg.

WHAT WENT WRONG:

Root causes (contributing factors/pre-conditions):

- Deviations from procedures/practices.
- Inadequate discipline – violation of safety rules/policies.
- Gaps in design – by-pass of the crown savers can be used for wrong purpose; second saver not working independent from first saver.
- Improper supervision – condone inappropriate behaviour of workers.
- Lack of risk assessment.
- Incomplete wire rope inspection method described in the existing procedure.
- Insufficient involvement of local management to foster right behaviour.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- If the travelling block or any other overhead equipment strikes the mast or crown block, then immediately STOP WORK and notify the Rig Move manager. The work shall not be restarted until technical checks have been made on the mast, wire rope, crown block and travelling block.
- Make sure that "Crown Saver" (installed on the drawworks) is set properly. Check visual and functional daily all crown savers installed on the rig.
- Your CCTV and Digilog system must be functional. You must report it immediately if it is not working. It is strictly forbidden to switch-off any of them!
- Do not take 'shortcuts' follow working procedures at all times.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate supervision

EUROPE ONSHORE

DATE: **Aug 13 2021**
COUNTRY: **Romania**
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

Around 10.15 am during the milling of the cement plug in the 51/2 column head, after about 20 cm of milling, a strong jet of gas and about 2-300 litres of crude oil and brine was released uncontrolled. The propelled gas jet hit an ACF (Agregat de cimentare/Pump truck) operator in the vicinity of the well cellar. Following the blow, he fell to the ground, hitting his hand and the head area.

The 112 service was called, and a fire truck and an ambulance and a police crew soon reached the well. IP was investigated on the site, then he was taken by ambulance and transported to Hospital for further investigations.

WHAT WENT WRONG:

Root causes/ basic causes:

- Inadequate hazard identification or risk assessment
- Inadequate supervision
- Inadequate training/competence
- Inadequate work standards/procedures:
- Inadequate/defective guards or protective barriers
- Inadequate/defective personal protective equipment

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always wear safety glasses, violating this rule will lead to consequences.
- While milling an abandoned well, always use the stripper and/or coil tubing.
- Never perform well abandonment operations without being aware of all the well data (reports, drawings).
- Analyze the entire operation risks and consider using equipment which reduces the risk to As Low As Reasonably Practicable.
- Fit for purpose and certified well control equipment is mandatory to be used during all operations including abandonment operations.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

EUROPE OFFSHORE

DATE: **May 29 2021**
COUNTRY: **Netherlands**
FUNCTION: None
CAUSE: Dropped objects
ACTIVITY: Unspecified - other
RULE: Unspecified

NARRATIVE:

Top deck: a scaffold plank has been found between V-1133 and the railing, posing a potential fall hazard.

WHAT WENT WRONG:

Not available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: **Jun 27 2021**
COUNTRY: **Netherlands**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Other issue

NARRATIVE:

The valve had been tested onshore by a Contractor, and had passed the required pressure test. However when tested offshore it failed the pressure test.

There is a discrepancy with the ID number of the valve in relation to the corresponding certificates.

There has now been a second valve sent to the rig, which has also failed the offshore pressure test.

WHAT WENT WRONG:

- Management: failure to implement or share best practice, through lessons learned, or from industry guidance or regulatory requirement.
- The Contractor Technical Work Instructions for drillpipe safety valves are not in line with API Standards and Specifications. It should consider that drillpipe safety valves are well control equipment. Also, increased understanding of workshop personnel on the usage of the equipment will result in increased quality and reliability of the equipment. For example, the wording "should seal at 250 psi and shall seal at 800 psi" could cause confusion, and may be replaced by "shall seal at 250 psi and shall seal at 800 psi". Furthermore, it should be emphasized that it is not allowed to first create a seal with a high pressure/flow ("bumping or seating" the valve), for a subsequent low-pressure test. After updating procedures, Contractor workshop hands should accurately follow written technical work instructions.

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- Equipment: inadequate maintenance or inspection in accordance with an approved maintenance or inspection regime.
- API Specification 7-1 and API Standard 53 are not aligned with regards to pressure testing safety Valves and Inside BOP's. This equipment is all manufactured according to API Spec 7-1 and hydrostatic testing is described in this specification, but not as detailed in API Standard 53. The difference in the hydrostatic testing procedure gives room for error and misinterpretation, especially with the low-pressure test.
- The difference in Class 1 (surface only) and Class 2 (Surface and downhole) safety valves should be included in the procedures of the Contractor as per API 7-1 table 3 service class definitions. A Class 1 valve is to be used at surface and is not intended to be run in the well. A Class 1 safety valve is not designed, tested and classified to hold pressure from external to internal. Only Class 2 safety valves shall be used to be stripped in the well.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- In this investigation it is concluded, after disassembling one of the valves, that the remaining force of the internal springs of the drillpipe safety valve was insufficient to properly close the valve at low pressure/flowrate.
- The valve however passed a workshop pressure test prior mobilisation because of the different pressure testing procedure/method applied in this workshop.
- Multiple discrepancies have been found between API 53, API 7, NS2 and the Contractor's internal (global) testing & maintenance standards/protocols. These differences allow for misinterpretation and unsafe usage of drillpipe safety valves. Therefore, it is recommended that the Contractor improves their global standards/procedures for maintenance and pressure testing of drillpipe safety valves.
- Also, further investigation on alignment between API 7-1, NS2 and API standard 53 could improve industry safety, preventing similar events from happening on other locations and resulting in serious well control incidents.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Jan 28 2021
COUNTRY: Netherlands
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

When moving a used drum with a capacity of 80 litres of coolant using an approved barrel clamp, the bottom of the drum ended up on the railing during the lifting work. Because the weight of the load decreased as a result, the barrel clamp became detached from the drum. The drum with a total weight of 85 kg fell into the sea.

After filling the diesel engine with coolant (24-01-2021), there were still 2 drums of coolant (1x full and 1 x +/- 80 litre) on the cellar deck of the platform. These drums would be moved from the cell deck to the mezzanine deck using the crane. This is the Deck where the chemicals store is located.

These activities were discussed before the lifting activities by means of a toolbox meeting. The tap check had already been done.

The communication between the crane driver and banksman was good and the orders given with regard to the lifting orders were clearly received and followed up.

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WHAT WENT WRONG:

- Operating equipment: incorrect tool(s), equipment or machinery handling, operation or selection.
- Management: failure to implement or share best practice, through lessons learned, or from industry guidance or regulatory requirement.
- Equipment: unknown, undetermined or unmerited equipment failure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review the situations in which these clamps are used to transfer drums and risk assess activities before commencing.
- Consider alternative, more secure methods to transfer drums.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: **Jun 5 2021**
COUNTRY: **Netherlands**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

During lifting work with East crane, the main hoist block came against the protective strip of the west crane boom while moving a tank. Crane driver immediately reported this and the broken strip was immediately removed. Other protective strip and boom on the West crane were checked; only minor paint damage. Main lifting block on the East crane checked, no damage.

WHAT WENT WRONG:

Operating Equipment: obstructed or restricted workplace/hazard inducing conditions.

Laydown area is very close to the boom of the west crane when viewed in height.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

No Causal Factors Allocated

EUROPE OFFSHORE

DATE: Jun 24 2021
COUNTRY: Netherlands
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Unspecified - other
RULE: Line of fire

NARRATIVE:

Heavy corroded Unistrut was only hanging on its cable, creating the potential for a dropped object above the stairwell on the northeast side top deck.

WHAT WENT WRONG:

Corrosion.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Feb 10 2021
COUNTRY: Norway
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

A cuttings hose was to be dismantled before a rig move. The hose was lifted by use of a deck crane and fibre lifting sling. When the hose-end was lifted up, the crane held the load in place and the IP entered line of fire, outside the dropped object protection cage to remove the security strap between hose and hose saddle (hang off point). With no warning the fibre lifting sling between crane hook and hose snapped. The hose fell down and hit the IP in shoulder area. The IP fell forward and hit the railing. Operations stopped, first aid team and support teams called to the area. The IP was transported to the rig hospital and onshore by SAR. IP was conscious but with pain in shoulder/chest area.

WHAT WENT WRONG:

Underlying causes:

- Inadequate job safety preparation (No SJA, inadequate permit to work and pre job meeting) leading to inadequate risk identification and risk reducing measures
- Inadequate governing documentation/procedure
- Inadequate design of saddle, cuttings hose and protection cage
- Work not according to lifting and pipe handling operations in the Safety Standard (e.g. operation not identified as blind lifting)
- Work in "Line of fire"

EUROPE OFFSHORE

ROOT CAUSE:

- The risk for handling hose for cuttings and the placement of the saddle and the protection cage, was brought up several times during construction, but the solution for storage and handling was accepted and not corrected.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure compliance to Safety Standard Job safety preparation for planning and performance of all jobs.
- Ensure lifts are performed according to Lifting and pipe handling operations in the Safety Standard.
- Ensure procedures are made for required activities/operations.
- Identify other activities where work is performed under hanging load, and take required actions.
- Consider design of equipment being lifted and surrounding structure/equipment with regard to possibility for hooking load, and take required actions.
- Ensure knowledge of and compliance to Life-saving rules.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: Dec 22 2021
COUNTRY: Norway
FUNCTION: Drilling
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

During reactivation of the BOP control system, a coupling on the 5000 psi shear ram boost system, came loose.

Hydraulic oil under 5000 psi (345 bar) pressure sprayed out from the 20 mm supply piping from the 5000 psi accumulator bank. This oil mist activated the smoke detectors, which then triggered a general alarm on the platform.

Two persons were present in the room when the incident happened. They were standing around 1.5 metres from the leakage point.

Both evacuated immediately through the alternative escape route into the neighbouring cement room. None of them were injured, but both underwent routine checks by the medic since having inhaled hydraulic oil mist.

WHAT WENT WRONG:

Root cause(1) - Design - Interface with the operator

Root cause(2) - Procedures and control of operations

Root cause(3) - Training - Lack of knowledge

Root cause(4) - Work place conditions leading to human errors

EUROPE OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Rec (1): Evaluate further actions to correct errors on shear ram boost unit, or equivalent solutions, in order to get the BOP back to operational.

Rec (2): Communicate the HSE alert and lessons learned to the Company globally through the HSE function.

Rec(3): Communicate incident report and learnings to fabrication vendor.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: Apr 21 2021
COUNTRY: Norway
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

The ongoing operation was running in the hole with 12 ¼" BHA. After a drill stand connection, the driller gave the command to open the slips and the slips opened; then the drilling stand was lowered down to be ran into the hole. When the lower tool joint went across the slips, a side of it touched dies segments causing the slips to be set in close position. The slips was clearly not fully open, however this needs to be confirmed by further investigation. When the slips suddenly went to closed position during RIH, the choked drill pipe caused the upper tool joint to hit heavily the torque wrench guide funnel, which parted and fell down on driller cabin roof, bouncing on the glass protection and falling on drill floor deck. The weight of the falling object is 44 kg. The height of the fall is the full drilling stand (approximately 28 m from drill floor). No injury of personnel, nobody was inside the red zone area.

WHAT WENT WRONG:

The driller did not visually verify that the PS16 was fully open and start running in hole with the drill string when still in opening sequence.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Torque wrench guide funnel dropped because the tool joint touched the dies segment of the power slips, causing the slips to be set in closed positions. The choked drill pipe on the power slips caused the upper tool joint to come in contact and heavily hit the torque wrench guide funnel, that parted and fell down.

The driller did not visually verify that the power slips was fully open and started running in hole with the drill string when the power slips was still in opening sequence.

RECOMMENDATIONS:

- Improvement on design of power slips to be investigated with manufacturer.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

EUROPE OFFSHORE

DATE: **Aug 19 2021**
COUNTRY: **Norway**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

The deck crew reported dropped objects found during night shift at 02:00 hrs. The two falling objects belonged to the turbine exhaust cover. One piece was found on deck in the loading area approximately 10 metres below the exhaust cover. The second piece was found approximately 6.75 metres below. The weight of each part was 3.5 kg. Fall energy 1st piece is estimated to 343 joule. Fall energy 2nd piece is estimated to 232 joule. A 3rd piece of the clamp was also found later on deck; 3.5 kg down 10 metres from the exhaust pipe.

WHAT WENT WRONG:

Defective equipment, tools or materials.

Inadequate planning and follow up of maintenance.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Exhaust cover fell down due to the locking clips being used was not suitable for the purpose. The covers was not regularly inspected and wear and tear to the exhaust cover locking clips was not identified.

RECOMMENDATIONS:

- Thorough visual inspection by RAT team and update DROPS plan scope and interval.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: **Dec 9 2021**
COUNTRY: **Norway**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

The rig was waiting on weather due to strong wind. The wind was measured over 60 knots from 150° when the incident took place. The rig is equipped with a Maritime Hydraulics drilling package. This includes Gantry Cranes for tubular handling. On the aft deck (pipe deck) there is a gantry crane for horizontal handling of tubulars like drill pipe and tools. On the front deck there is a gantry crane for vertical handling of riser joints. Both cranes have an identical crane cabin equipped with AC unit (Air Condition) on top of the cabin. The AC unit is stored inside a box with dual hatches. Hatches are horizontal when closed, and vertical when opened. Hatches are for maintenance access for the AC unit

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inside. As weather picked up the day before, the operation eventually came to a stop due to the weather conditions. Both wind and sea state became too strong to continue the operation. Around 11:00 hrs a person noticed a hatch lying in the track for the gantry crane when going down from the drill floor. All activities were stopped in the area for TOFS, area was barriered off immediately, OIM was informed, all personnel and client informed, carried out thorough check of the area and all similar equipment around the rig. Additional securing on remaining hatch put in place. The hatch was found on the starboard side next to the tracks. A total drop of 6 metres and a travel distance of about 10 metres. The wind directions match the travel path of the hatch. The investigation team have not been able to pinpoint the exact point of contact when the hatch fell but the hatch got a deep dent that indicates an impact with a sharp structure. This again can imply an impact with the Gantry Crane track below the cabin.

WHAT WENT WRONG:

Strong wind and rust on one of the hinges.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Risk of AC unit hatch hinges being corroded, and hatch falling down, was not identified. Need for maintenance was not identified due to the access to inspect the hatches being difficult because the handrail was missing.

DROPS inspection not conducted according to plan.

RECOMMENDATIONS:

- Identify and secure similar equipment.
- Install handrails where needed to do necessary inspections.
- Update DROPS inspection plan.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: Jan 17 2021
COUNTRY: Norway
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

After a container had been lifted in place, the precursor of the crane was slackened, leaving the forerunner of the 4-leg sling end in a small arc down on the catwalk. The load handler freed the hook and communicated "free hook – hoist" to the crane operator. The crane operator confirmed "free hook – hoist" and lifted gently. The hook became stuck in a small guide that was sticking out of the railings on the catwalk. The function of the guide is to prevent the pipes from falling between the railings and the feed mechanism itself on the catwalk. When the crane wire was tightened, the hook came loose from the guide where it had become stuck. The crane operator felt a small "jerk" in the crane as he saw the load handler get hit and disappear out of sight.

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WHAT WENT WRONG:

- The incident shows that even small local adaptions of facilities can introduce new hazards. Such hazards must be uncovered through risk identification and risk management through the change process.
- The management system does not clarify and emphasise the risks in the final stages of lifting operations.
- At the time of the incident, insufficient attention was paid to the hook during the final stages of the lifting operations.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure that sufficient attention is paid also during the final stages of lifting operations.
- Ensure that modifications to the drilling areas are carried out in accordance with the Company's governing documents.
- Review crane and lift work processes and requirements with a view to clarifying and emphasising the risks in the final stages of lifting operations.
- The risks associated with a crane lift are not over until the hook is released and lifted in a safe manner, thus ensuring that the hook cannot hit personnel or get caught on anything.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: May 13 2021
COUNTRY: Norway
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Construction, commissioning, decommissioning
RULE: Safe mechanical lifting

NARRATIVE:

During the demolition of a chemical ramp, an access ramp part had to be lifted. The work team consisted of 1 crane driver and 3 deck operators. It was an asymmetrical crane lift, with unknown centre of gravity, and a weight of approximately 1.5 tons. The structure to be lifted was secured to the main ramp structure by 2 load straps and connected to the deck crane using 2 off 2-leg lifting slings.

When loosening the first load strap, the object started rotating and one person was squeezed against the remaining chemical ramp.

WHAT WENT WRONG:

The team's limited experience with this type of asymmetrical lift, unclear management of the lifts and unclear coordination. of roles and tasks in the work team are, in the investigation team's assessment, human factors that affected the team's planning, risk assessment and quality of the toolbox talks. This led to an increased likelihood of operational and human error. Weaknesses in the quality of planning, risk assessment and toolbox talks led to a lack

EUROPE OFFSHORE

of common understanding in the team of the individual steps in the operation, the risk factors in each step, and how these risk factors were to be handled. As a result:

- The crane operator and signaller were not aware that tension from the crane hook was transmitted to the tension in the upper load strap.
- The work team was not sufficiently aware of how the object could move.
- The situation with blind lifts was not identified during planning and risk assessment.
- It was not communicated within the work team who was to loosen the load straps.
- The injured person had their attention on other tasks before the accident and her understanding of the situation when moving towards the object, into an unsafe zone, just before the accident, was unclear.
- The work team was not sufficiently aware of the object's centre of gravity when attachment points for the lifting tool were chosen.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always keep your back free, keep distance to load and lifting operations, and do not expose yourself to Line of Fire.
- Be aware of the interaction of the different roles in the work team.
- The responsible line management must identify lifting activities that may cause challenges, such as lifting asymmetrical objects, and ensure that a thorough risk assessment has been performed.
- The entire work team must participate at the work place for the lifting object during the risk assessment of the lifting operation.
- Using the opportunity to learn across project and operations when assessing risk and planning lifting operations requiring complex rigging.
- Self-evaluation and verification of compliance with the work process for lifting operations with offshore cranes.
- Clarify the concept of Lifting special loads.
- Review and improve checklist for planning and risk assessment of lifting operations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

EUROPE OFFSHORE

DATE: Jun 9 2021
COUNTRY: UK
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Other issue

NARRATIVE:

Whilst dredging, a conductor unexpectedly toppled over vertically but away from a 2-person dive team working in the area. No injury to the divers.

WHAT WENT WRONG:

Mechanical cut completion data missing from drawing. Asset site specific information not up to date.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Asset specific site information document created for each site; to be updated post intervention.
- Work Pack review process strengthened.
- Resourcing reviewed to ensure adequate competence for future campaigns.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: Jun 12 2021
COUNTRY: UK
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Other issue

NARRATIVE:

Failure of starboard recovery winch fairlead assembly.

WHAT WENT WRONG:

On initial review of the area, it was clear that the stbd recovery winch fairlead assembly had failed and come free from the deck, along with the fairlead failure the stbd recovery wire had parted. It is presumed that, following the fairlead failure, the wire contacted the kick plate causing the wire to part.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Towed units are prone to Yaw.
- The construction and installation of a temporary unit should be able to withstand all potential forces it supports. The forward laydown area should have been barriered off to all personnel. The winch should have been set to pay out above a set load.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: **Jul 23 2021**
COUNTRY: **Kurdistan Region Of Iraq**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Unspecified

NARRATIVE:

While pulling the drill pipe from the hole, the floorman operating the power tong had broken the first connection on the heavyweight drill pipe and the other two crew members racked it back then they went and stood next to the driller side air winch, away from the rotary table.

The tong operator was lifting the power tong to move it away from the heavyweight drill pipe when the tong started to drop towards the rig floor. At the same time the wire rope from which the tong had been suspended, started to free fall to the rig floor. The floorman operating the power tong ran between the draw-works and drillers console to safety. The other two crew members standing next to the drillers side air winch (Assistant Driller and Floorman) ran to the safety of the doghouse.

Approximately 90 metres of wire rope fell 50 metres to the rig floor. The weight of wire rope is estimated at 130kgs.

No one was injured and no damage was caused to any other equipment because of this incident.

WHAT WENT WRONG:

Work Planning, Control, and Administration

1. There was no job planning done to identify the tools/ work practices required and control the risks involved in the critical job. The crew were told to do the rig up of equipment without specific instructions and adequate supervision.
2. Inspections are carried out to meet schedule targets rather than identifying and correcting deficiencies.
3. In general, lack of administrative support is evident with case-to-case management of operational requirements rather than having a systematic and organized approach and commitment from contractor Management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Use the off-driller's side tugger line to suspend the power tong, as the immediate corrective action implemented after the incident.
2. A SOP/Work Instruction to be developed to provide instructions, as per LOLER standards, to crew for proper termination using bulldog grips. Procedure should describe as a minimum: Sizing of grips, Spacing and turnback requirements, Torque specifications. This will be developed with the assistance of a lifting specialist.
3. A Torque Wrench must be available and used for making wire rope terminations and make up torque referenced to manufactures recommendations
4. Supervisors to be made aware of their accountability for monitoring and verification of job completed by the crew as per SOP.
5. All crew to be trained and certified by a competent trainer, as per LOLER standards, on wire rope terminations using bulldog grips.
6. Power tong suspension assembly to be included in the dropped object inspection check list for rig floor and to be inspected by Assistant Driller and validated by Rig Manager.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

MIDDLE EAST ONSHORE

DATE: **Nov 30 2021**
COUNTRY: **Kurdistan Region Of Iraq**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Unspecified

NARRATIVE:

The operation was to lower the mast. The Mast ultimately relies upon a single PLC (programmable logic controller) supported by a single UPS (uninterruptible power supply). The UPS suffered a momentary failure, perhaps only in the order of milliseconds, but of sufficient duration to disconnect the PLC.

As a result, the drawworks winch VFD (variable frequency drive) unit lost its controlling input and power to the motors disconnected. With a derrick weighing 55 tonnes cantilevered at a distance of 45m out from the superstructure, the momentum caused the drawworks to free-wheel for one second until the emergency brakes activated to stop the drawworks drum from rotating. However, the supporting winch cable has sufficient stretch to permit the derrick to impact the supporting cradle after the drum had stopped. After the cable contracted, the derrick came to rest some 100mm above the cradle. The descent had been successfully halted.

WHAT WENT WRONG:

Undetermined. No apparent evidence as to why the UPS failed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Replace the UPS with two units with longer battery autonomy and of ruggedised, industrial standard rather than commercial standard given the rigors of the application in this drilling environment.
2. Consider replacing the PLC with a dual redundant CPU and I/O cards.
3. Ensure that the PLC has an active historian function for post event data analysis.
4. Perform and record daily checks on the UPS to make sure that it is working and that the batteries are healthy and charged as this unit not only supports control of the mast raise and lower but all drilling functions on the rig.
5. Ensure that UPS check is included in the SOP for mast raising and lowering activity.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

MIDDLE EAST ONSHORE

DATE: **Dec 15 2021**
COUNTRY: **Kurdistan Region Of Iraq**
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

RIG Draw works did not respond to the joystick manual stop command and continued to pull the travelling block into the crown block (crowned out).

Four protection systems are installed to prevent crown out: 3 systems failed due to power failure to the control circuit and 1 system was activated but it was unable to stop the drawworks. All 4 systems were checked during rigging up and before the incident and were working correctly at that time.

Root cause of the above failures was a short circuit which burned a 24V DC transformer that powers the PLC which receives the emergency stop signals. The PLC forwards the emergency stop signals to the various devices which need to be shut down. Because of the failure of the 24V DC transformer, the PLC was not powered up and it didn't transmit the signals.

WHAT WENT WRONG:

- Physical or Physiological Stress.
- Other Self-Imposed Stress, Rig was off operational rate and there was possibly a perceived sense of rush.
- Inadequate practice.
- Electrician failed to act on damaged wires in junction box in VFD.
- Inadequate communication/implementation of policy/procedure/practice.
- Failure to properly resolve issues with short circuit that caused the burn out of a 24V DC transformer that powers the PLC which receives and sends messages to the Drillers control panel (Joystick) and also controls the emergency stop signals from the Drillers control panel.
- Improper production incentive.
- Rig was off operational rate and there was possibly a perceived sense of rush.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Replace all damaged cables in VFD Control junction box.
- Parameters to be changed to allow ESD to function correctly on DW AC motors.
- Replace Emergency push button with a new one.
- Install a relay on ABB drives to kill ABB Drives when VFD Kill switch is activated.
- Install fuses on the PLC output in the VFD PLC Cabinet.
- Calibration procedure should be posted at the Driller's cabin and calibration tests shall be performed by the Rig Manager, Tool pusher Driller and Chief Electrician.
- Prior to any start-up of routine work Driller and Tool Pusher shall check calibration of travelling block ESD height.
- Work orders (PM System) for the Draw works, AC Motors and Travelling block anti contact device and Emergency Stop buttons.
- Inspection of Travelling Block and crown to be carried out by third party inspectors.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: Aug 4 2021
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

While the Roustabout was moving around the cellar grating, he stood on the cellar hatch which dislodged allowing him to fall through the gap into the cellar a fall of approximately 3.5m deep. He was not injured.

WHAT WENT WRONG:

- Inadequate update/refresher training – buddy system should have been reinforced during pre-tower and weekly safety meetings.
- Inadequate coaching – Assistant Driller was not available to coach/supervise work activities.
- Inadequate delegation – the was not an alternate supervisor nominated after the Assistant Driller was not available.
- Inadequate management information – Roustabouts unclear as to whom should be giving them work instructions.
- Inadequate monitoring of construction/fabrication/assembly – cellar grating design needs to be a consultation between civils, drilling, and production.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. A more permanent solution should have been used to mitigate the hazard of the partially opened grating as the cellar pump was used almost every day. This had had become common practice as there grating had been barricaded.
2. Drilling contractor needs to implement a policy that formally nominates a senior person to supervise Roustabouts (green hats) when the Assistant Driller is unavailable. This is to be communicated to all Drilling contractor persons onsite.
3. Company well cellar gratings need to incorporate small openings for cellar pump hoses to mitigate the need to remove grating.
4. Future cellar gratings should be welded so that the grating clamps cannot be removed. The grating should be re-designed to allow for easy removal for drilling spool changeout, but also not have small sections of grating that are easily removed when grating clamps are removed, as these can loosen and deteriorate over a period of time.
5. Drilling contractor needs to remind all employees of the buddy system. This needs to be communicated at HSE Weekly meetings and pre-tower meetings.
6. Drilling contractor do not have SOPs for routine activities such as cleaning the cellar area.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Nov 23 2021
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Office, warehouse, accommodation, catering
RULE: Other issue

NARRATIVE:

A fire broke out in the Camp Kitchen at the rig Main Camp. All personnel were accounted for. There were no injuries reported.

The source of the fire was a deep fat fryer that, 10 minutes previously, had been used to cook French fries for the Rig Crew's mid-day meal. The investigation was not able to determine whether the deep fat fryer electric heating system was switched off after use, or whether it was isolated and an electrical fault occurred that allowed the electric heating element to continue to heat the cooking oil.

WHAT WENT WRONG:

- Inadequate initial instruction/ skill training – No documented evidence that HSE Training has been carried out for Camp Supervisor, Deputy Camp Supervisor or Catering personnel.
- Inadequate initial instruction/ skill training – No documented evidence that a "Standard for Kitchen Operation" was in place, which included the use of Deep Fat Fryers and Cooking Oil Safety and Hygiene.
- Inadequate practice – Use of fire extinguisher, drills – (No evidence of recent Fire or Emergency Response Drills for the Rig Camp or Kitchen facility.
- Inadequate communication/ implementation of policy/ procedure/ practice - failure to change the cooking oil every 3 days as per Corrective Action from previous incident.
- Condone misuse of equipment/tool – Camp Supervisor and Deputy failed to ensure cooking oil was changed as required.
- Condone improper/inappropriate behaviour – Chef continued to use contaminated cooking oil.
- Inadequate monitoring/closure of audit actions - Camp Supervisor and Deputy failed to ensure implementation of the previous incident agreed actions (regular changing of the cooking oil).
- Inadequate shelf life/Validation of re-use of materials/Equipment.
- No pre issue checks were carried out on the fryer prior to sending to the rig site. Was issued with one side of the fryer defective.
- Inadequate assessment of preventative maintenance needs - No routine maintenance scheduled all maintenance is reactive.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Provide training for relevant personnel in correct preparation, use, cleaning, and maintenance of all kitchen equipment with the potential to cause harm, loss, or damage.
- Fatigue management must be considered with preparing shift rosters. Camp supervisor to review if there is a requirement to change shift patterns to enable kitchen staff to have sufficient rest periods.
- Company to develop, train and implement a "Kitchen Operation HSE Standard" which aligns with the requirements of ISO-22000 (Food Safety Management Systems), and which includes the use of Deep Fat Fryers and Cooking Oil Safety and Hygiene. "Standard for Kitchen Operation" shall include Fire Risk Assessment and control measures including AFD, Alarms, First Aid Firefighting equipment and media and pre-determined attendance for supporting additional personnel and FF extinguishing media from Rig etc. NB and future catering contracts.
- Review Catering Management System with particular focus on: Catering Management System ownership and clearly defined roles and responsibilities for Company and Contractor personnel in accordance with existing Contracts. Records to be maintained in terms of:
  - Catering Equipment condition and planned preventive maintenance.
  - Catering Equipment fault reporting, replacement part orders, parts replacement, and re-commissioning.
  - Catering Facility Inspections, remedial actions and close out (to be reviewed at Weekly HSE Meeting).
  - Camp Boss/Deputy Camp Boss and Catering Staff Job Descriptions, Role Responsibilities and Role Specific Induction and Training.
  - Requirement for Catering Crew and Camp Boss to hold relevant food handling and Hygiene Certifications.
  - Cooking oil changes.
- Clarify Catering Contract performance management review requirements and establish regular communication lines. Camp Supervisor or Deputy to attend morning Contractors Meeting to hear, first hand, comments on catering quality and to discuss maintenance issues.
- Review Emergency Response Arrangements at Rig Camp, including but not limited to: Establish and post a Weekly Roster of ERT Cover for Rig Camp, Posting of Station Bills in all areas of the operation, carry our Emergency Response Drills including Muster Drills. All Rig sites.
- Contractor HSE to oversee compliance with Cooking Oil Safety and Hygiene procedure and formally report non-compliance to Camp Supervisor with immediate copy to Company contract holder (Operations Superintendent) or his designate.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST ONSHORE

DATE: Feb 7 2021
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Office, warehouse, accommodation, catering
RULE: Bypassing safety controls

NARRATIVE:

Asset Damage HiPo - Contractor camp electrical fire - an electrical fire started from cabin RG-1 distribution board while a person was sleeping. The fire started in the junction box on the interior of the cabin. There was no smoke detector in the cabin but the occupant awoke due to the noise and smell. They extinguished the fire and raised the alarm.

WHAT WENT WRONG:

Sub-standard electrical equipment and procedures were used during the manufacturing of the cabin. No RCD or AFCI was used. No fire proof cabling was used. No sleeves were used to protect the wiring when passing through the wall panels.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Follow Emergency Procedure for FIRE.
2. Ensure rented equipment are of the correct standard and requirements as to be set out by OSV.
3. Ensure electrical inspection regime is established and performed on monthly basis.
4. Do equipment inspections and acceptance checks before equipment is to be used.
5. Set a minimum standard requirement for any equipment to be used on site, before supplied
6. Ensure all safety equipment is fitted to the cabins, even if rented and do regular inspections to confirm functionality.
7. Establish emergency lighting outside the cabins, not only inside, to allow visibility during darkness.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: Sep 24 2021
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

MIDDLE EAST ONSHORE

NARRATIVE:

NGL hose burst.

WHAT WENT WRONG:

NGL Hose burst at NGL loading Bay.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

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CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: Nov 21 2021
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Construction
CAUSE: Unspecified - Other
ACTIVITY: Office, warehouse, accommodation, catering
RULE: Other issue

NARRATIVE:

Security guard noticed a fire in a construction contractor laundry container. Fire alarm activated and ensured the laundry container was electrically isolated. Although the alarm was sounded no one proceeded to muster, instead people either participated in attempting to fight the fire or were observing those that were attempting to fight the fire. Emergency Response Team arrived on location and extinguished the fire. Company personnel instructed contractor personnel to muster and carryout a POB check to ensure all personnel are accounted for. The fire was extinguished after 10 minutes of water application. No personnel injury occurred. The cause of the fire incident was dryer unit lint filter not being cleaned.

WHAT WENT WRONG:

- Inadequate experience: Unexperienced staff hired with no competency assessment.
- Inadequate initial instruction/skill training: No training matrix No on job training were delivered to the laundry crew.
- Inadequate orientation/induction.
- Inadequate initial training.
- Inadequate inspection/repair/maintenance: PW failed to identify lint and put a maintenance program in place to clean the lint filters.
- Inadequate assessment of needs and risks.
- Inadequate assessment of needs and risks.
- Inadequate product/service planning.
- Inadequate risk identification/evaluation in development of standard.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor to undertake a fire risk assessment then update the risk register accordingly with laundry dryer fire risks and the mitigations covered within the risk register.
- Contractor review the training matrix and update accordingly to ensure all competency requirements to undertake their allocated task are fully captured.
- Ensure a weekly maintenance service program and a safety inspection checklist is in place for laundry, dryers lint filters are cleaned on weekly basis. An assurance audit schedule involving Contractor camp boss and HSE oversight is also required to be put in place.
- An exercise and drills matrix are to be developed; these should be conducted on weekly basis. Drill reports must be recorded and documented and available for review.
- Contractor supervisors are to be trained in emergency response management and practised during in simulation exercises.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: May 22 2021
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Unspecified
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Energy isolation

NARRATIVE:

Major grass fires within perimeter fence.

WHAT WENT WRONG:

Major grass fires occurred within the perimeter fence, caused by ignition from an old power cable exposed on the ground connected to one of the OPF towers. The fire rapidly spread due to high winds, but was successfully extinguished, without any damage to Plant and facilities.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

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CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

MIDDLE EAST ONSHORE

DATE: **Jan 27 2021**
COUNTRY: **Kuwait**
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
RULE: Working at height

NARRATIVE:

A foreman of a maintenance contractor fell from the ladder while coming down from scaffold platform. He was working with his crew for spading a flare gas line. He suffered a broken vertebra (back bone) as a result of the fall and has undergone surgery.

WHAT WENT WRONG:

Not available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: **Apr 3 2021**
COUNTRY: **Oman**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well services
RULE: Working at height

NARRATIVE:

Derrick man fell from 1.6 metres while he was assisting the driller to install 8" poorboy degasser valve, at the same time the valve fell down hitting his helmet and his head. The IP was examined by company medic and transferred him to nearest hospital for further checks.

WHAT WENT WRONG:

- Derrick Man right index finger has limited movement due to previous home injury. However, he didn't report this.
- Driller did not set the proper example to the people working under his supervision by involving himself in the task instead of supervising the task.
- Derrick Man and Driller normalized the risk and didn't use the right work at height equipment.
- The valve design was correct and is as per OEM, however, the design can be further improved to avoid potential fall.
- Proper equipment (Platform ladder or MEWP) was not used for the task.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Crew to complete Working At Height training as per the Training Matrix. Awareness & Practical (Specialized 3rd Party).
- HSE Self-Assessment to be conducted for all supervisors to identify the area of improvement and actions to be taken accordingly.

MIDDLE EAST ONSHORE

- Initiate a "CCTV Project" for live streaming people behaviour on Rig Floor & provide coaching/mentoring based on feedback.
- Apply a better Engineering Design for the Valve to avoid potential fall.
- Conduct Engagement session workshops with all supervisors by including Planning, Communication, Supervision, Leadership, and Intervention & Empowerment to Stop.
- Disciplinary action to be applied for the Driller for not following the procedure.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Feb 9 2021
COUNTRY: Oman
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

A brackish tanker was returning back from production station, suddenly the lock of the fifth wheel broke and disconnected from the king pin which caused the prime mover to disconnect from the tanker trailer. Nobody was injured.

WHAT WENT WRONG:

- The whole locking mechanism of the fifth wheel system of UD type of prime movers is very weak. It is very easy to get broken and no proper retention in place.
- Annual Inspection issue: Third party inspection isn't detailing what had been inspected and what were the findings (Detailed MPI was not provided).
- Rough road condition.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Performed awareness session for all drivers in the weekly meeting highlighting this incident.
- Repaired this tanker by replacing the fifth wheel with MAN type.
- Performed a survey to see the tankers which are having the same type of fifth wheel (5 tankers in total).
- Instructed the owners of other 5 tankers to replace their fifth wheel with MAN type.
- Detach the prime mover from the tanker at least one time per month to perform visual inspection.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: **Sep 10 2021**
COUNTRY: **Oman**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

A contractor brackish water tanker discounted from prime mover while the driver decided to use a shortcut road (lumpy road) to enter the well location, which lead to the tanker trailer tipping over.

WHAT WENT WRONG:

1. The base of the king pin was not fixed with 8 nails according to the accepted technical standards, but was fixed with only 6 nails.
2. No visual inspection for king pin bolts by third party company.
3. The driver used the shortcut unpaved road with a high speed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Visual and practical monthly king pin & fifth wheel inspection for all prime mover with must disconnected and connected during the inspection.
2. Informed all the contract owners that must and mandatory to send the trucks to the third party to inspect the vehicle to issue a technical certificate.
3. Share the incident with the drivers for awareness.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation intentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST ONSHORE

DATE: Jul 8 2021
COUNTRY: Oman
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Construction, commissioning, decommissioning
RULE: Driving

NARRATIVE:

A water tanker driver was driving up the ramp to the well pad location; while reaching the top of the ramp, the driver stopped on the steep ramp due to activities going at the entrance of the well pad. Due to a slippery surface, the tanker moved backwards. The driver applied the brakes trying to control the tanker but no success and the tanker tipped over. The driver's right Wrist was fractured.

WHAT WENT WRONG:

- Inadequate identification of worksite/job hazards.
- Not following company approved design of ramp due to constraint in well location space.
- Absence of heavy vehicle periodic maintenance and inspection checks.
- No Risk Assessment Conducted for such Locations.
- Unavailability of banksman to manage traffic.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The steep inclination of the ramp should be avoided while designing ramp.
- Drivers should be trained to identify workplace hazards as a proactive measure to avoid accident situations.
- A banksman must be available during heavy vehicle movements in such locations.
- Water Tanker Inspection and proper maintenance is a must to avoid leakage.
- Frequent verification on driver awareness of Hazard Identification process.
- Effective implementation of STOP Work Authority policy when observing unsafe acts/conditions.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

MIDDLE EAST ONSHORE

DATE: Aug 1 2021
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Production operations
RULE: Work authorization

NARRATIVE:

Utility area substation received a trip alarm from 33KV/66KV back-up transformer. A rover operator was dispatched to check the area and observed a fire and black smoke coming from the transformer. The emergency response initiated, and the fire was extinguished with the application of foam and dry powder by support of RLIC.

WHAT WENT WRONG:

The transformer was designed to be operated in a grounded system and for continuous operation, however it was not operated as stated in the design specification (i.e., eliminating the neutral phase (ground)).

The change of the operating philosophy before the commissioning of the transformers was not managed properly. The hazard of operating the transformers without neutral phase and any potential stress on the system/bushing was not identified. With the change, there was also no detection of potential earth fault.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

For the new project handover, the check of the design specification and operating philosophy consistency need to be involved in the project handover process.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Jul 25 2021
COUNTRY: UAE
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While Running in Hole (RIH) with Heavy Weight Drill Pipe (HWDP), the Derrickman was using winch with a chain to pull a stand from the finger board to latch it into the TDS elevator. At that time, the pin of the two-part shackle got caught on the chain, causing its release and drop of the hook, from Monkey board to rig Floor.

Shackle landed approximately 1.5 metres from the Directional Driller, who was on the rig floor near the mouse hole.

No injury or damage to equipment.

Hook weight approximately 0.4 Kg. Height from where hook dropped was approximately +/- 90 Feet (30 metres).

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

Improper Use of Equipment/Inadequate Equipment:

- The Hook and the Chain was not part of the original design of the Monkey board winch.

Improper Decision Making/Lack of Judgement

- No permit or justification given for using two-part shackles to connect the chain with the winch line.
- Derrickman failed to identify that the two-part shackle was loose.

Servicing of Equipment in Operation

- Derrick logbook does not indicate any details of chain/hook being placed at monkey board.
- DROPS survey did not cover the details of the winch and how it is secured.
- Weekly DROPS Inspection does not specify verifications of the Monkey board winch and associated elements.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Conduct a survey on all land rigs to verify if a similar hook, chain and two-part shackle were used at the monkey board. Take out of service and replace chain hook assembly with wire rope slings and small hook for safer practice.

Replace all two-part shackles which are used at the derrick with four-part shackles.

Conduct DROPS survey on the monkey board, to include the monkey board winch securing.

Update weekly DROPS checklist to include the Monkey board winch verifications, as recommended by the drops survey.

DROPS Awareness Session to be conducted with all rig crews.

Apply painting on the Rig floor with DROPS colours: Green, Red and No-Go Zone.

Ensure adequate crew change scheduling with consideration of fatigue and stress.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: Dec 9 2021
COUNTRY: UAE
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

After the successive failure of the 2 pilots on the morning, unburnt sour gas was emitted from the tip of the flare stack.

An hour later, toxic gas detectors in well pad area (400m away from the flare stack) initiated an Emergency Shut Down-3 (ESD-3).

MIDDLE EAST ONSHORE

Shortly afterward, H2S detectors were triggered farther than 100m away.

Another ESD-3 was initiated in another area, 1h and 20 min later.

Volume released 40 kg of gas with 2 kg of H2S during 150 min.

WHAT WENT WRONG:

Flare pilots 1 and 2 went off and the auto-ignition system did not manage to re-ignite the flow of gas.

Ignited pilot is visually difficult to see during day time. Failure to recognise pilot status/Alarm.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Operating Procedures revision and associated documentation is necessary to ensure they are up to date and no missing information.
2. Implementation of detector system is necessary: Consider installation of mobile H2S Toxic Gas Detectors and Alarms and procurement of Thermal Camera to visualize non-visible flame.
3. Inspection Frequency of Flare Tip and Pilots FG system using Drone camera and routine Nitrogen flushing should be increased.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST OFFSHORE

DATE: **Aug 8 2021**
COUNTRY: **Kurdistan Region Of Iraq**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

Pressure/flow surge in upstream process systems.

WHAT WENT WRONG:

Extreme kinetic energy, caused by a pressure/flow surge in the upstream process systems, resulted in the displacement of LPG 16" pipework, 6" LPG Pumps suction pipework and LPG Product Pumps from their fixed positions. It was fortunate that no loss of primary containment occurred during the event.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

-

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: **Jul 17 2021**
COUNTRY: **Qatar**
FUNCTION: Drilling
CAUSE: Cut, puncture, scrape
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

At the stairway of platform top-deck, 2 persons were removing the gravity fill up hose from a chiksan connection. After the hose has been removed from the connection, a tag line was used as a guide while the hose was being transferred/lowered to platform top deck by air tugger. The tagline was short, so the crew released it to continue lowering the hose.

When the tag line was released, the hose started to swing. The IP was standing in the direction of the swinging hose and the hose end fitting hit the IP on helmet and caused laceration on right side of head.

WHAT WENT WRONG:

- Failed to use Stop Work Authority.
- Failed to follow procedure (No PTW and not part of Daily activity List).
- Lack of Supervision.
- Lack of training .
- Hazard not identified - Wrong choice of rig tugger, which caused pendulum effect while lifting the hose.
- Failed to follow Management of change - Rig jacking JSA to be updated with potential to lift equipment onto the platform.

MIDDLE EAST OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Regular Stop Work Authority exercises to be conducted on the rig.
- Supervisors to be reminded that any job planned for the day ahead has to be documented on the Rig Daily Activity list.
- Permit to work training to be carried out.
- Update jacking JSA Lift plans.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Jul 31 2021
COUNTRY: Qatar
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

While attempting to run in hole with drill string for circulating at bottom, the Injured Person (along with three co-workers approached the rotary table to get ready to remove slips on the drill floor. The Driller started raising the Top Drive System, resulting in the pipe handler taking the weight of the drill string, as the TDS assembly was comprised of a saver sub and two Internal Blowout Preventers (valves had backed off at the top connection between the upper IBOP and the top drive quill shaft). Excessive load caused pipe handler damage, resulting in a metal pin dropping to the rig floor, striking the IP on his head and shoulder. The metal pin (weighing 7 kg) dropped from a height of approximately 27 metres. The IP sustained a laceration wound on his head (approximately 2.5cm) and an abrasion on the right shoulder (approximately 2 cm).

WHAT WENT WRONG:

The lock ring at the upper IBOP was not installed correctly, which resulted in the lock ring not functioning as designed to prevent excessive torque at the upper IBOP to quill shaft connection.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update PPM to include a weekly check of IBOP lock rings, as required by the manufacturer.
- The scribe mark on the IBOP assembly shall be visible at all times.
- Instruct appropriate personnel to verify IBOP assembly connections after ramp failures.
- Ensure TDS and pipe handler components are installed as per OEM requirement.
- Job Safety Analysis to be reviewed to include drops hazards, red zone management, and use of auto slips to prevent manual slip use when feasible.

MIDDLE EAST OFFSHORE

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: **May 25 2021**
COUNTRY: **Qatar**
FUNCTION: Drilling
CAUSE: Falls from height
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

The preparation for Load testing using MPC (Multipurpose Crane) was ongoing.

The operator was situated inside the basket of the MPC crane to configure basket and knuckle boom to position the basket to carry out operation. While moving the knuckle boom basket upwards, the basket got stuck in between pedestal and the beam. The operator stopped the Job and came out of the basket in order to use the remote operating MPC panel. Whilst operating from remote panel to free the basket, the basket (365 kg) got disengaged from the boom and fell from about 10 metre height onto the deck.

WHAT WENT WRONG:

- Improper management of change.
- Poor Job training.
- Lack of Hazard Identification.
- Weakness in manufacturer design.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Provide dedicated training from crane manufacturer.
- Design solution from crane manufacturer.
- Review work procedure.
- Permit to work and Hazard identification refresher training.
- Signage/MPC Manual available at the control panel.
- Dedicated parking area for access or/and egress of the MPC basket.
- MPC pre use checklist to be signed by Tool Pusher.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST OFFSHORE

DATE: Feb 11 2021
COUNTRY: Qatar
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

During the final cut of an 18 5/8" casing, the Injured Person (IP) was holding two steel slings that were attached to the casing stump as he prepared for the final cut penetration. At the completion of the cut, the stump and attached casing cutter 'jumped' as kinetic energy was released. The IP was struck by the casing stub and subsequently fell off the work platform. The IP sustained lacerations on the left frontal areas of the scalp and left upper eyebrow.

WHAT WENT WRONG:

- Too much tension was applied to the lifting tugger. No procedure/guideline is in place that states what the correct tension should be, or what the correct lifting arrangement should be for the task.
- The Job Safety Analysis (JSA) omits to mention that the casing should not be held onto. The IP was holding onto slings that were holding up the casing, which put him in the Line of Fire.
- The IP had no experience in cold cutting operations, but was instructed by his Supervisor to help out with the cold cutting operations.
- A Toolbox Talk was not performed, and the JSA was not reviewed with the IP before undertaking the task.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update the cold cutting method statement/procedure with site-specific procedures for rig-up to securely hold both ends of the pipe being cut.
- Note: Consider 'direct line of sight' lifting controls; tag lines to control the cut-off casing piece – keep personnel and hands away, and ensure correct tension is applied to the lifting appliance.
- Update the cold cutting JSA to capture the hazard of stored energy, highlight Line of Fire awareness, and include a caution for personnel to stay away from the casing being cut.
- If the job or conditions change, stop and take Time out for Safety (TOFS). No personnel to undertake a task, even for brief period for minor support, without a full JSA briefing.
- Run a focus campaign on Situational Awareness and the hazards of Stored Energy.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST OFFSHORE

DATE: **Jul 17 2021**
COUNTRY: **Qatar**
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Construction, commissioning, decommissioning
RULE: Energy isolation

NARRATIVE:

Fire from 400v electrical socket equipment at the process area: A scaffolder was passing close to the unit and smelled something burning. He came closer to the equipment and observed smoke and flames were coming from the electrical socket power supply to the Multi Sensing Unit. He immediately informed the control room and used a fire extinguisher to extinguish the fire. The technician switched off the power and unplug the cable at the platform socket.

WHAT WENT WRONG:

- Wrong cable and shoes sizes in the electrical plug.
- Equipment failure due to lack of pre-mobilization preparation and QC verification.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Demobilize all existing units and replace with fixed units.
- Inspect all existing units to ensure the correct cable type (green 16mm2 RFOU cable) and sockets wiring.
- Inform manufacturer electrician to service the equipment.
- Use Digital thermal camera to check the temperature of the connections for overheating.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: **Dec 14 2021**
COUNTRY: **Qatar**
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
RULE: Safe mechanical lifting

NARRATIVE:

Jacket level survey was ongoing by using a levelling tool installed on the top of the jacket pile and connected with grommets and shackles to the jacket legs on both sides.

Pins of the right assembly slipped out from their position causing the grommet, right side link plate and lower pin to fall down.

As a result of the event, the grommet remained on the scaffold being connected to jacket leg, and the link plate with the pin (weight approximately 600kg) fell into the sea and landed on the jacket mudmat.

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The area around the levelling tool was restricted with access to workers as per JSA control measures in order to do not have any personnel in the line of fire (LOF) during this activity and further level survey.

WHAT WENT WRONG:

- Levelling System Design (configuration with link plate, grommet, and shackles) didn't allow the perfect alignment of the link plate with shackle (presence of lateral forces).
- Engineering design of the system didn't consider the possibility of a system misalignment with the proposed configuration that could determine lateral force on safety pin.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Identified a proper engineering solution (replace the link plate and pin with 800t shackle) in order to rectify the inadequate design allowing the safe completion of the jacket operations.
- Reviewed the installation procedure, inserting a dedicated section concerning the alignment check before commencing operation and revised the drawing according to the modified configuration for performing the jacket levelling completion.
- Share the lesson learned to all project and areas, in particular to consider the system misalignments and the present of lateral forces.
- Consider the line of fire and the positions of the personnel during the whole process.
- Use the levelling system with link plates when directly connected to the pad eyes. If not possible, the revised solution with the shackle in place to the link plates should be preferred as the shackle would have sustained portion of horizontal load.
- Revise the safety pin design of the levelling tool equipment in order to account any potential lateral forces. Welding of washers/bolts on pin shall not be considered as possible solution due to risk of cold cracks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: Feb 16 2021
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Falls from height
ACTIVITY: Construction, commissioning, decommissioning
RULE: Safe mechanical lifting

NARRATIVE:

During the launching of the Fast Rescue Craft (FRC), the crew used the onboard lowering system to extend the davit and to lower the FRC to the water as normal practice. The boat was lowered smoothly at a consistent descent rate to approximately 1.5-2.0m from the water surface. Suddenly the brake release wire jammed at the davit and would no longer pay out wire.

Since the IP applied his weight on the brake release, the FRC continued to be lowered and the IP had his hand extended with the handle up to 2 metres above the FRC deck. Once the FRC reached the sea level, the IP released the brake handle and fell to the FRC deck sustaining bruising to his left hip and left ankle.

MIDDLE EAST OFFSHORE

WHAT WENT WRONG:

- Management System (standard Policies for launching and recovery).
- Maintenance (cable sheave with kink on the brake wire not identified during spooling).
- Training/familiarization personnel (No quick reaction to the situation).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Replace the brake release wire and fall wire with approved and complete certificates.
- Inspect and certify all FRC mechanisms including sheaves wires and other parts by approved 3rd party.
- Monitoring and inspection of wire rope condition when spooling during launching and recovery to be included in Technological Risk assessments.
- Review and update maintenance procedure.
- Conduct System Awareness program onboard related to system/SOP Strict implementation.
- Maintain training, familiarization and refresher for the use of equipment including during emergency situations.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA ONSHORE

DATE: Oct 21 2021
COUNTRY: Canada
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

After switching over of different tank products and a period of stable operations plant conditions, a spill was detected coming from the overflow of a tank.

WHAT WENT WRONG:

Diluent tank overflow.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Evaluate how projects incorporate operating condition (and fluid characteristics) changes during HAZOPS and MOC.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

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DATE: Aug 6 2021
COUNTRY: Canada
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Work authorization

NARRATIVE:

A worker was tasked to cut out and replace a drain valve. The welder used a grinder to cut into a ¾" boiler feed water (BFW) line to remove the drain valve, subsequently the pipeline was breached and a stream of BFW was released. The worker did not receive any injury. All work was stopped. The line was fed by a charge pump, the pressure in the line was 12,500 KPA (1813 psi) and the temp at that time was 43C.

WHAT WENT WRONG:

1. Comm system needs improvement - A work order was removed from the LOTO by operations but this action was never communicated to the schedule team which then allowed the contractor to receive a work package to complete work on equipment was not isolated.
2. Scheduling needs improvement - Key personnel was not on-site to perform the outage.
3. Work package/permit needs improvement - The safe work permit did not include the work for the task the contractor was performing. It was for the work they performed for in the morning, the work scope was very generic for the task as well.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Implement formal and informal assurance activities, including a focus on safe work permitting, LOTO, control of work and hazard assessments.
2. Implement standard agenda item in Weekly Ops Leadership Meeting Terms of Reference - discussion hold point on upcoming outages that will include resourcing adequacy, schedule lock and LOTO development and readiness.
3. Request and complete external review of existing equipment and LOTO and blinding procedure. Review to include commentary on process complexity, relevance for ease of use and understanding at the site level.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: Apr 13 2021
COUNTRY: Canada
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Unspecified

NARRATIVE:

Operators noticed a blind laying on the ground in the high lift building of the direct river intake. Upon further investigation, they concluded that the 470 lb. blind had sheared off the high lift pump discharge header and fallen to the ground from a height of 20 feet, landing in a frequently travelled area. The blind was in the 12 o'clock position, although several other 20 inch blinds in the building were in the 9 o'clock position.

WHAT WENT WRONG:

Analysis identified that a small crack existed in the toe of one weld.

The blind was unsupported and subject to minor vibration for over 14 months of operation.

Piping Specification limits on Spec Blind weight were not followed. Standard states "Spectacle blinds shall be used in cases where the total weight of the fitting is less than 45 kg (100 lbs)" and that "in cases where the spectacle blind exceeds 45 kg (100 lbs), a spacer and blind paddle set shall be used".

A deviation to exceed the CLCH piping size limit (from 18 to 24 inches) was submitted and approved but did not include blinds. Piping Spec specifies that a spacer and paddle blind be used for sizes greater than 8- inches.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Engineering to communicate spectacle blind weight/size limits to teams.
2. Share company-wide alert regarding spectacle blind weight/size limits with recommendation to identify and remove non-complaint blinds. Spec blinds over frequently travelled walkways should also be identified and considered for removal.
3. Remove all spec blinds from hazard locations at the direct river intake.
4. Major Projects team will review this incident and reinforce material selection requirements.

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CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Feb 8 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

The DSR was alerted that one of the SPM crew members had injured his left middle finger. The IP was attempting to adjust a joint of panic line by rolling it with his hands from the edge of the forklift's right fork back onto the pipe trailer. As the panic line joint rolled off, the IP's left middle finger became caught between the moving panic line joint and the joint of panic line that was already on the trailer. All required PPE was being worn.

WHAT WENT WRONG:

- Inadequate work oversight or enforcement of work standards.
- Training exists but individual was not trained.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Oct 7 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

The crew was in the process of Nippling Up the Blow Out Preventer (BOP). The BOP was raised from the horizontal to the vertical position with the BOP Wrangler and was being maneuvered over the well centre. As the BOP was being moved toward the well centre, the Motorman observed a potential line of fire contact hazard between the handle on the lower pipe ram and the choke line that was laid over the off-driller side substructure. The Motorman and

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Floorhand (injured employee) stepped forward into the substructure to move the choke line out of the line of fire. The Floorhand placed his right hand on the substructure prior to reaching down to move the choke line. As the Floorhand placed his right hand on the substructure the top of his right wrist was struck by a 36-inch pipe wrench. The contact resulted in a contusion to the top of his right wrist. The Floorhand was treated at the rig with first aid measures.

WHAT WENT WRONG:

Root cause: Procedures and control of operations - Lack of procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Drilling contractor will create a formal pre-lift inspection for BOP and BOP Wrangler.
- Drilling contractor will revise Nipple Up Critical Task Assessment to include use of formal BOP and BOP Wrangler inspection.
- BOP Wrangler will be equipped with octagon shaped sticker to indicate inspection requirement prior to raising.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: Jun 9 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While running a 3-1/2 inch pipe (liner joint), a section needed to be set down due to thread galling damage while torquing the connection. A lifting device (nubbin) was attached to the liner joint's top-end (box), and the liner joint was hoisted into the air. As the liner joint was raised, team members were moving the liner joint's bottom-end (pin) towards the catwalk when the nubbin disengaged from the liner joint's box-end. The liner joint fell onto the power tongs causing them to strike a team member in the right-hip as he was leaving the rig floor. The liner joint continued to fall until it came to rest with one end (pin) on the rig floor and the other end (box) laying on the ground.

WHAT WENT WRONG:

Liner joints ran with incorrect storage compound on threads and damaged threads.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Proper thread compound is applied prior to makeup.
2. Ensure threads in good condition.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: Aug 14 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Exposure electrical
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While starting drying shaker #2, the motor man stepped onto the grating walkway and experienced a shock to the back of his neck. At the same time there was a spark on the grating. The Motorman called for assistance to verify that the power cable had been de-energized. The breaker was confirmed in the tripped position. After inspection of the wire, it was found that the red hot wire (270V) had been pierced.

WHAT WENT WRONG:

Inadequate installation, Lack of Quality Control From Provider, Failure to Recognize a Hazard.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Communicate to Third Party Service Providers to perform quality control measures on equipment before delivery. Rig down and remove equipment from location.
- Communicate the need to inspect equipment after installation and/or repairs. Held conversation with management for pit cleaners about hazard recognition and awareness.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: Jul 7 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While replacing the sand conveyor on a fracking operation, the conveyor operator, with a designated spotter, backed the unit up and then began folding the conveyor's arm to its resting position. At that point, the spotter was called away. The spotter stated he gave the operator a STOP hand motion and left the area, however, the operator continued to fold the conveyor arm and contacted the overhead powerline. The operator received a shock and was injured.

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WHAT WENT WRONG:

Conveyor operator contacted energized overhead powerline.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Identify and mark overhead powerlines
- Reinforce spotter expectations

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: Mar 5 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While attempting to transfer fuel, a cam-lock connection failed and injured a contract worker. When the pump was engaged, a misaligned valve directed the fuel to a closed ball valve. The transfer hose became over-pressured causing a cam-lock connection to fail. The failed connection struck the Injured Persons's hard hat; he was also impacted by the energy release.

WHAT WENT WRONG:

Misaligned valve directed pressured and fuel to a closed ball valve.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Review vendor equipment for pressure relief systems.
2. Hose securement and ensure exclusion zones are established.
3. Walk the line - confirm valve alignments.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

NORTH AMERICA ONSHORE

DATE: Feb 6 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Slips and trips (at same height)
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

After stepping out of a catering trailer, the IP stepped down to the ground and began to walk from the rear of the trailer directly towards their personal vehicle, approximately 25 feet away. While walking across the location, the IP tripped on the corner of a line-backer and fell to the ground. No one witnessed the event; however, other personnel on location noticed the IP lying on the ground within moments. Upon checking on the IP, the IP stated that they struck their knee while falling and wished to be transported to the hospital.

WHAT WENT WRONG:

HES Procedures or Safe Work Practices not available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Feb 11 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Other issue

NARRATIVE:

During rig move operations and while loading Crane counterweights (stacked) onto a flatbed trailer, a Truck Swamper and Crane Rigger were attempting to guide the stacked counterweights onto the trailer. Both employees were on the truck bed using their hands to try and position the load. The crane rigger had his left index finger positioned between the two weights. The two weights were pinned together and, when suspended, there is a gap between the weights. The Truck Swamper truck swamper flagged the Pole Truck operator to lower the load with the winch line catching the Crane Rigger's finger between the two weights. The IP jerked his hand from his glove realizing he had a hurt finger.

WHAT WENT WRONG:

- HES Procedures or Safe Work Practices inadequate.
- Inadequate work oversight or enforcement of work standards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA ONSHORE

DATE: Jun 15 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Unspecified

NARRATIVE:

A packer (plug) was being run in a well during plug and abandon (P&A) operations when the well began to flow. The well was shut in and secured using the Blowout Preventer (BOP), however, leaks in the production casing allowed liquid and gas to be released at the surface from the cellar. The rig crew opened the well to an open-top tank to divert the flow away from the immediate rig area. The well continued to flow until it was brought under control using heavier weight fluid.

WHAT WENT WRONG:

A leak from a hole in the surface casing allowed fluid and gas to be released.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure well handover process is followed.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: Dec 28 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Unspecified

NARRATIVE:

Rig crew was picking up and running tubing. While raising the joint, the Driller tried to apply the brake but couldn't push the mechanical brake handle down. The Driller informed floor personnel the brakes were not functioning and, unsure of the problem, engaged the emergency stop. A team member guided the tail (lower) end of the joint behind the handrail and personnel exited the rig floor as the blocks continued to descend. The elevators, bails and blocks came to rest on the handrail and the joint bent to a 90° angle. No injuries occurred.

WHAT WENT WRONG:

Brake linkage bolt caught on bent walkway bracket.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Inspections do not cover the brake assembly path of travel.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

NORTH AMERICA ONSHORE

DATE: Jun 9 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

After making up the mud motor, the Driller was raising the blocks to pick up a collar. While raising the blocks in 4th gear, the Driller attempted to shift down but unintentionally shifted up causing the blocks to accelerate upward. Upon seeing the blocks accelerate, the Driller engaged the brake and disengaged the clutch, but the upward momentum of the blocks carried them through the Crown-o-Matic set point and into the crown. The drill line parted, and the blocks fell within their guide rails to the rig floor. No personnel were injured during the event.

WHAT WENT WRONG:

Unintentionally shift up of blocks causing the blocks to accelerate uncontrolled upward.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Document Crown/Floor Saver set points and tests.
2. Installed gear shift in an easy to operate location.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: Apr 13 2021
COUNTRY: USA
FUNCTION: Production
CAUSE: Explosion
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

During start-up of a new facility the team was filling a water tank when it exploded. The resulting pool fire caused secondary explosions in another water tank and oil tank. Of the people onsite, only minor first-aid cases occurred.

WHAT WENT WRONG:

Ignition source was determined to be either a static discharge inside the water tank or flashback through the low-pressure flare system.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Ensure adequate grounding and bonding.
2. Ensure flame arrestors are installed at an appropriate distance from flare tip.
3. Ensure all equipment is purged of air prior to facility start up and potential ignition sources eliminated.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

NORTH AMERICA ONSHORE

DATE: **Jun 18 2021**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Production operations
RULE: Energy isolation

NARRATIVE:

An ESP Technician reached into the panel and unplugged a serial cable from the engineering port on PIC card. The bare ground wire on the 120 V circuit feeding the PLC came in contact with the energized fuse in the drive causing an arc flash.

WHAT WENT WRONG:

- Original Job scope required a LSA EI which was completed properly (Hazard recognition). After returning energy to the panel, the IPs tolerated the energy hazard and failed to install the proper safeguard (Arc Flash PPE/ Energy Isolation).
- PIC & Work Authorization breakdown. The Company Representative was the original PIC and delegated the WA properly. Later the PIC duties were given to another contractor without electrical expertise. Work Authority cannot be delegated to contractors.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Work Authorization responsibilities should be transferred to competent site leadership during shift change and/ or job handover.
- Re-opening electrically charged equipment requires re-authorization, PPE, and energy isolation where zero energy is demonstrated.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation intentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

NORTH AMERICA ONSHORE

DATE: **Jul 22 2021**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

While a contract crew was opening the manway to clean an oil storage tank, a flash fire occurred, burning two workers.

WHAT WENT WRONG:

Non-intrinsically safe tool utilized within hazardous area.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure non-intrinsically safe tools are not utilized within hazardous area.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: **Oct 6 2021**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Maintenance, inspection, testing
RULE: Work authorization

NARRATIVE:

After draining water and hydrocarbon emulsion from a heater-treater using system pressure to push the fluids through a hose to a vacuum truck, the truck operator disconnected the hose from the heater-treater. Fluid remaining in the hose and from the truck released out of the open-ended hose and sprayed the heater-treater resulting in a flash fire. The vacuum truck operator received burns.

WHAT WENT WRONG:

Excess fluid and gas from hose ignited by nearby gas burner which was not turned off.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Turn off or isolate all nearby ignition sources when draining heater treaters.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

NORTH AMERICA ONSHORE

DATE: Feb 2 2021
COUNTRY: USA
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Construction, commissioning, decommissioning
RULE: Working at height

NARRATIVE:

IP was standing on well cellar grating when the crew started picking up the grating with the backhoe. The crew's first lift was hooked to the grating which underlaps the rest of the grating causing the others to collapse within the cellar.

WHAT WENT WRONG:

- Design did not consider human factors.
- Operations Procedures did not exist.
- Inadequate work oversight or enforcement of work standards

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: Feb 1 2021
COUNTRY: USA
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
RULE: Working at height

NARRATIVE:

Employee was descending a platform using a stationary ladder. The Employee lost his footing and fell 6'.

The employee sustained fractures in his leg.

WHAT WENT WRONG:

Not available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

NONE: None: Unspecified

NORTH AMERICA ONSHORE

DATE: **Jun 3 2021**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While transferring sand into sand silo transport, the operator was monitoring sand levels and observed that his bulk trailer compartment ruptured during offloading operations. The operator immediately suspended air supply and disengaged the PTO system. Operations were immediately suspended and notifications made to PIC of incident. There were no injuries reported.

WHAT WENT WRONG:

Preventive Maintenance, Inspection, Testing or Repair – less than adequate

1. Recent weld repairs were inadequate to prevent the hull from rupturing. (Pending Metallurgy Test). The new PRV PSI rating higher than maximum allowed pressure inside the trailer's hull.
2. Although the manufacturer recommends integrity test after modifications, damage or repair to pressure related system and components; however, there are no required equipment maintenance standards to mandate annual or periodic integrity test.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Action Items:

1. All sand haulers will have PRVs with a PSI rating lower than the PSI rating for their trailers' sand compartments, per test pressure.
2. All sand hauling equipment will be operational, free of mechanical defects and disrepair.
3. Reinforce the use of PAUSE and standard operating procedures for sand delivery with accountability at all levels.
4. Re-assigning Sub-Contractors and other trucking from transport or routes for this Frac Contract locations and facilities.
5. Team will review all applicable wellsite and company processes with vendors upon arrival.
6. Resource/Support reallocated for process auditing and contractor management.
7. Critical checklist for onsite validations by Sand Coordinators/Supply Chain. Pre-Task checks will be performed, the equipment will be inspected prior to commencing in offloading procedure.
8. Explore requirement for any "aftermarket" repair/modification to equipment that is designed for use of pressure related application; will have verification and certification of test.
9. Update/develop Pre-Mob/Load Out Checklist for requirements and inspections. To be approved and delivered for sub-contractors to utilize during Pre-Trip Inspections, prior to deployment.
10. Sub-Contractor/High Risk forum annually with all Fracking Sub-Contractors.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

NORTH AMERICA ONSHORE

DATE: **Mar 18 2021**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Unspecified - other
RULE: Line of fire

NARRATIVE:

At approximately 1:20 AM, a worker was walking toward the washcar located at the Base of Feed and was struck by a light duty vehicle travelling 13 km/hr. The worker fell face first to the ground at which point the front left wheel rolled over the worker's right ankle resulting in a fractured leg. The worker then rolled over onto their back and saw the rear left side wheel travelling towards their left leg and body. The worker was able to push against the truck using their left leg to avoid further injury.

WHAT WENT WRONG:

- Inadequate lighting southeast of washcar.
- Truck pillar and side mirror reduces visibility of driver.
- IP and light vehicle driver were focused on their perceived highest risk resulting in inattentional blindness (known as Salience bias in Safe Choice).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Reassess risk of pedestrian/vehicle interaction across the site and determine if additional controls are necessary.
- Refresh training, procedures, and communication of Working Around Mobile Equipment requirements and applicability to Light Vehicles, including design, construction and speed limits.
- Define walking paths.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: **May 6 2021**
COUNTRY: **USA**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

NARRATIVE:

While performing emissions upgrade work on the facility, a contractor employee was manoeuvring a telescopic manlift and came into contact with a 12.4 KV overhead powerline. This resulted in an arc flash event and parting of the line damaging the manlift.

WHAT WENT WRONG:

Manlift contacted by powerline.

NORTH AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Identify and mark overhead powerlines.
2. Reinforce spotter expectations.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: Feb 12 2021
COUNTRY: USA
FUNCTION: Unspecified
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Line of fire

NARRATIVE:

Around 10:45am, the M/V was performing diesel transfer operations. An all SWA was called, and the operation was suspended because weather conditions were deteriorating (increase in loop current). The IP disconnected the fuel hose and secured it to the crane hook. As the hose was being lifted by the west crane, the hose got caught on a tote tank which prompted the crane operator to slack off the lift. While attempting to free the hose, the IP's hand was caught between the lifting apparatus of the hose and the tote tank. The hose then parted at the breakaway coupling from the applied stress to the hose and approximately three gallons of diesel was released overboard while retrieving the remaining section of hose and securing it back onto the facility. The NRC was notified.

WHAT WENT WRONG:

Weather/Wildfire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

NORTH AMERICA OFFSHORE

DATE: Jul 15 2021
COUNTRY: Mexico
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

When installing the ecological tray in the BOP area, one of the support tubes rolled and fell from a height of 6 metres above the CTU, and subsequently fell on the main deck of the HCP at a height of 3 metres.

WHAT WENT WRONG:

1. The JHA recommendations were not followed.
2. The tubular supports did not have an adequate design.
3. The equipment does not have accessories for securing tools or materials at height.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Replace the current ecological tray for equipment for waste containment with a manufacturer's certificate.
2. Leadership training for supervisors (OIM, ITP, TP, Driller, Crane Operator, Crew Chief, Ship Engineer, Assistant Crane Operator, Assistant Driller) by qualified and approved personnel.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Oct 16 2021
COUNTRY: Mexico
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Safe mechanical lifting

NARRATIVE:

While lifting a Heavy weight with a hydraulic lifter, the HW slipped and bounced, hitting the upper part of the catwalk ramp. No injuries or damage were reported..

NORTH AMERICA OFFSHORE

WHAT WENT WRONG:

1. Not compliant with specific procedure to perform the task.
2. The JHA does not include all the required steps that are necessary for the activity.
3. The measurement of the load to be lifted (3 ½" HWDP) was not verified in comparison with the range of the insert to be used.
4. The stop the job policy was not applied.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Develop a training assurance program.
2. Communication campaign reinforcing stop the job policy.
3. Communicate the findings of this event.
4. Leadership skills development plan for senior and middle management personnel.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Aug 12 2021
COUNTRY: Mexico
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well services
RULE: Bypassing safety controls

NARRATIVE:

While removing a hanger ball from the rig floor, it got stuck. While a worker was trying to release it, the tool suddenly moved and the hanger ball rotated near the worker causing no damage to person or facility.

WHAT WENT WRONG:

1. Job Safety Analysis (AST) recommendation not followed.
2. The personnel (Tool Pusher) was positioned in the line of fire (tubing hanger) to pull the retained.
3. The stop the job policy was not applied.
4. Using a short pup joint under the hanger ball difficulties handling at ground level.

NORTH AMERICA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Improve follow up of training program.
2. Reinforce the stop the job policy and line of fire.
3. Carry out a performance evaluation for contractor personnel.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Jan 30 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

While in the process of cleaning a clogged auger leading from the big bag hopper to the mixing hopper, the IP used an 18" pipe wrench to remove the plug. After the plug was removed he observed a build-up of chemical material and debris. He used the pipe wrench hook to dislodge the chemical build-up, and while holding the hook end of the pipe wrench his hand slipped causing his finger to make contact with the hopper auger, severing the distal tip of the finger.

WHAT WENT WRONG:

- Design did not anticipate the conditions.
- Design did not consider human factors.
- HES Procedures or Safe Work Practices not utilized.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA OFFSHORE

DATE: Sep 29 2021
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well services
RULE: Unspecified

NARRATIVE:

In the process of rigging up a 4" swivel to the east side permanent piping on the +50, a worker utilized a 36"x1" valve bar weighing 11.5 lbs to gain a mechanical advantage and rotate the swivel. While rotating the swivel, the worker lost their grip on the bar, and it bounced through a 2" deck perforation and descended 31' to a lower deck. There were no personnel in the area of the dropped object. However, the area was not barricaded below. DROPS calculator potential outcome is Fatality.

WHAT WENT WRONG:

Personnel executing the task were working on a permanent platform deck level (i.e. not on a ladder or scaffold) and did not consider themselves to be working at height.

- Personnel executing the task did not identify, during pre-job planning or toolbox talk, a gap that existed in the deck that was large enough for the valve bar to fall through to decks below.
- Valve bar did not have a tool tether.
- Task specific work instruction did not identify dropped object exposures and prevention.
- Pre-job toolbox talk did not identify dropped object exposures as the gap was considered too small for an object to fall through.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lessons Learnt:

- The controlling DROPs program and training was not clear on what constitutes tasks that involve working at height with regard to working on elevated platforms/decks.
- Deck opening and leading deck edges where not adequately assessed as part of the pre-job dropped object risk assessment.

Recommendations:

- Consider implementing a standard site specific dropped object policy when working on elevated decks, that would classify the work as "working at height" with associated controls applied (i.e. tethered tools, exclusion zones).
- Site leaders should assess where it is possible to install permanent pipework in a an effort to reduce the use of temporary pipework.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

NORTH AMERICA OFFSHORE

DATE: **Jan 14 2021**
COUNTRY: **USA**
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

A 70 lb beam clamp, 110 lb chain-fall, and associated rigging was dropped when moved along the I-beam during removal process. One crew member was working from a rope access supported by two crew members on the deck, inside the barricaded drop zone.

WHAT WENT WRONG:

Failure to tie-off the beam clamp prior to loosening the clamping bolting and moving it along the I-beam resulted in it becoming disengaged from the I-beam and falling to the deck below; landing within 12 - 17 feet from the crew members.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Rope access work should have a stand-alone risk assessment and job plan.
- Workers should be kept out of drop zones whenever a dropped object hazard is present.
- Speaking up and stopping the job is a critical line of defence.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

RUSSIA & CENTRAL ASIA ONSHORE

DATE: Sep 24 2021
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Unspecified - other
RULE: Other issue

NARRATIVE:

Due to the heavy rains, the shore of the river slid where the pipeline was passing.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Regular site inspection has to be put in place.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: Dec 15 2021
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Unspecified - other
RULE: Unspecified

NARRATIVE:

The part of the tank side was forced down due to the high pressure inside the tank, no damage to facility.

WHAT WENT WRONG:

Outdated tank side, lack of inspection, overpressure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Proper maintenance, more inspections are required.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: Jan 29 2021
COUNTRY: Kazakhstan
FUNCTION: Construction
CAUSE: Cut, puncture, scrape
ACTIVITY: Construction, commissioning, decommissioning
RULE: Line of fire

RUSSIA & CENTRAL ASIA ONSHORE

NARRATIVE:

A carpenter working on a concrete form sustained a laceration to the small finger on their right hand while working with circular saw. The injury required amputation of the injured finger.

WHAT WENT WRONG:

- HES Procedures or Safe Work Practices not utilized.
- Hazard not recognized.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: Jan 28 2021
COUNTRY: Kazakhstan
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Maintenance, inspection, testing
RULE: Working at height

NARRATIVE:

During the pre-inspection of flow lines from production well, the IP approached the crawler tractor to collect materials from the body of the equipment when the operator initiated a turn to the right, striking the person

WHAT WENT WRONG:

- Mistake or mental slip.
- HES Procedures or Safe Work Practices did not exist.
- HES Procedures or Safe Work Practices not utilized.
- Hazard not recognized.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: **Jan 16 2021**
COUNTRY: **Azerbaijan**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

During the dismantling of the crane, the crane's balance was disturbed and the crane was tilted to the side of the pipeline as a result of the slab falling into the sea.

WHAT WENT WRONG:

The lifting procedures were not followed correctly and risks were not considered enough.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Work planning needs to be done correctly, competency of the supervisors is to be increased, all risks have to be taken into account, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: **Mar 10 2021**
COUNTRY: **Azerbaijan**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

The lifting equipment fell on the pipe in the area, but no damage was detected.

WHAT WENT WRONG:

In order to release the riveted right drill pipes from the well, the lifting truck rolled over the right side as a result of shaking created by the difference in forces.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Correct analysis has to be conducted prior to the work.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

SOUTH & CENTRAL AMERICA ONSHORE

DATE: **Jan 19 2021**
COUNTRY: **Argentina**
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well services
RULE: Bypassing safety controls

NARRATIVE:

During BHA leak test, a 3300 psi release occurred in wellhead. The damage was reported only over equipment.

WHAT WENT WRONG:

1. It was not possible to capture weak signals, when working with directional tools.
2. The organization failed to identify the need for a Shallow Test procedure.
3. There is no approved procedure for performing Shallow Tests.
4. The closed master valve does not allow to read pressure in the annular space in the manometre/choke manifold panel.
5. It was not verified that the master valve was closed, to carry out the correct venting.
6. The pressure relief set in the Amphion system did NOT manage to avoid the sudden increase in pressure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Implementation of an "alarm" system to warn about open manual master valve and closed HCR. In addition, "BOP closed" signage and record of the use of annular BOP during drilling.
2. Follow up to identify improvements in the team's tools or equipment in such a way that possible slippage is avoided at the time of wedge setting.
3. Prepare and deploy the procedure for this type of operation.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Mar 15 2021
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

During anchor setting activities, operators used a manual tool to hold the torque and ensure its setting. At some point the manual tool released its tension and struck 4 workers in the line of fire.

WHAT WENT WRONG:

- Improper use of manual tool/procedures: manual tools are not allowed for anchor setting operations, as stated in the procedures. If it is needed it must be authorized.
- Inadequate supervision: anchor setting operations are critical tasks that must be supervised by the Company Man, who was not there at the time of the incident.
- Failure to warn hazard: the anchor was being over-torqued and no one acknowledged the accumulation of energy and potential release.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Identify critical tasks during operations and apply the controls needed to ensure safe operations (Control of Work), including the necessary supervision.
- Ensure written procedures of critical tasks remain updated, accessible and understood by every worker.
- Implement activities to enhance operational discipline and organizational culture.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: Dec 10 2021
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Bypassing safety controls

NARRATIVE:

After a programmed plant shutdown and during start up activities, problems with fuel gas pressure caused several Plant shutdowns. Pressure protections in a well pad where disabled and HIPPS system was degraded after its first closure. This led to losing every layer of protection for over pressure scenarios in the well pad, causing an overpressure in the main pipeline, with no LOPC. As the pipeline reached 1.6 of its maximum operating pressure, it is considered a High Potential Event.

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WHAT WENT WRONG:

- Bypassing safety controls: wellhead valves had been forced open for a manoeuvre and hadn't been checked for the start up. They were bypassed at the time of the event. The HIPPS system valves were damaged after their first activation. They were bypassed at the time of the event.
- Inadequate practice: the override system used to force open the well-head valves usually remained set on the valve and was activated when necessary, without a proper registration/authorization.
- Failure to warn of hazard/Lack of sense of vulnerability: the activation of the HIPPS system should have trigger a deeper analysis and should have warned operating personnel about the bypassed well head valves. People in the control room did not notice the high pressure alarm.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Include every activity in the oilfield when planning a Plan Shutdown.
- Authorize, register and communicate every SCE bypass.
- Ensure every operator knows and understands the different layers of protection of the process and the impact of bypassing them.
- Implement activities to enhance operational discipline and organizational culture.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: Feb 16 2021
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

During personnel transportation, passengers identified signs of somnolent driving. They immediately asked the driver to stop and rest.

WHAT WENT WRONG:

- Non-compliance with procedures/contract: drivers did not rest and did not have the required amount of hours of sleep as the procedures establish.
- Inadequate supervision: there were no controls in place before each ride. The company did not fill the risk assessment document for authorization. There was no evidence of safety audits or performance evaluations.
- Drivers did not know or apply the activity suspension policy.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Include periodic audits and performance reviews of critical services and enforce their compliance.
- Verify drivers' rest the required amount of hours.
- Implement technological devices in order to identify somnolent driving.
- Verify everyone knows and is responsible for implementing the job suspension policy.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Oct 25 2021
COUNTRY: Bolivia
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Unspecified - other
RULE: Other issue

NARRATIVE:

It was reported in the external parking area, outside the area of operations a lateral overturning of a 190 ton crane during the replacement of a left rear tyre, an activity carried out on the crane operator's own initiative without due authorization, without a risk analysis or Work Permit, and there was no communication of the activity to the Company Man and/or HSE supervisor. At the time of the incident there were 3 people in the area and 2 drivers were in the cab of their vehicles parked near the crane truck, who were not affected by the incident.

First Aid attention is reported to have been given to the crane operator, who was operating the unit at the time of the event.

Considerable material damage was reported to the Crane, Slick Line unit truck, and Wire Line unit truck, due to the crane boom falling down over these vehicles.

WHAT WENT WRONG:

Root causes:

1. Maintenance management.
2. Procedures and control of operations - Lack of Permit to work .
3. Training - Lack of risk perception.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Perform training about lifting load procedures (Company Procedure "lifting and hoisting loads" and Bolivia procedure "Equipment and elements for lifting loads and use of cranes and forklifts").
- Clearly establish lifting requirements and communicate them to the contractor. This must be included in the induction presentation to all contracting companies.
- Reinforce the preventive observations program and the stop authority in D&C activities
- Require the Contractor to have an updated HSE training and education program for crane operators and handlers and establish a follow-up to identify compliance.

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- Include a specific training requirement about lifting risk assessment.
- Specify in the bridging document the areas of responsibility for hazard control management. (Essential for the management of work permits.)

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: May 16 2021
COUNTRY: Ecuador
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport - Land
RULE: Driving

NARRATIVE:

Upon taking a curve, the vehicle was impacted head-on by a third party vehicle traveling in the opposite direction.

WHAT WENT WRONG:

Third party vehicle driver lack of competency. Newly paved road, with gravel and no signage.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improve third party driver's driving behaviour. Manage road's improvement conditions. Improve driving management.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

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DATE: Oct 2 2021
COUNTRY: Ecuador
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

During bathymetry activities on a sub fluvial crossing, it was detected that a 45 metre segment of the pipeline was missing its external concrete coating.

WHAT WENT WRONG:

Lack of periodic updating of the identification of external factors and by third parties, within the risk studies.

Lack of permanent specific communication protocols with local governments to socialize the presence of the pipeline and its sensitivity to the action of third parties (tourist activities, mining, roads, among others).

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Establish communication protocols with local governments to publicize the presence of pipelines and mitigate possible impacts caused by third-party activities (tourism, mining, roads, among others).

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: Apr 23 2021
COUNTRY: Peru
FUNCTION: Production
CAUSE: Cut, puncture, scrape
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

While a worker was positioned below a 24" pipe performing a cutting task with a circular electric saw, the tool rebounded sharply, causing an injury to his right thigh.

WHAT WENT WRONG:

Personal Factors:

- Inadequate experience: Worker does not register experience in Excavation activities. Foreman does not have specific experience for the position of supervisor.
- Inadequate Initial Instruction/Skills Training. Foreman and workers were not trained in the specific work to be performed.
- Inadequate Discipline: Failure to comply with the procedures of "Excavation and Hazardous Energies Isolation" and "Control of Work".

Labour Factors:

- Tolerate deviation from policies/procedure/practice: Supervision tolerates deviation and does not comply with Excavation and Hazardous Energy Isolation and Control of Work procedures.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- We should not give a tool a different use for which it has been designed.
- The electric circular saw should not be used for cutting activities on uneven ground.
- The circular saw cutting tool should only be used on a flat base.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: **Sep 29 2021**
COUNTRY: **Peru**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Excavation, trenching, ground disturbance
RULE: Work authorization

NARRATIVE:

During excavation work in the NGL spheres area, operators removed the mechanical protection of a buried low-voltage power line, without letting the person in charge know about its existence. The operators impacted the cable´s protection when carrying out manual excavation tasks.

WHAT WENT WRONG:

- Inadequate experience: Workers do not register experience in Excavation activities.
- Inadequate Initial Instruction/Skills Training. Foreman and co-workers were not trained in the specific work to be performed.
- Inadequate Discipline: Failure to comply with the "Excavation and Hazardous Energies Isolation" and "Control of Work" procedures.
- Tolerate deviation from policies/procedure/practice: Supervision fails to comply with Excavation and Hazardous Energies Isolation and Control of Work procedures.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure and monitor the process of Control of Work training and authorization

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

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DATE: **Jul 10 2021**
COUNTRY: **Peru**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

During normal operation of the Fractionation Plant, turbogenerator No. 2 tripped, causing disturbance in the electrical generation system which lead to a black out.

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WHAT WENT WRONG:

Labour Factors:

Inadequate Product/Service Design:

- Inadequate assessment of needs and risk. (Lack of study/simulation for PMS shedding settings).
- Inadequate validation of the service/product design. (Lack of performance testing and load shedding program).
- Inadequate process design/standard/specification criteria. (Low frequency and high rotating power scenario was not identified.)

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Make identification of hazards and risk assessments visible and ensure they are taken into account as from the Design, Engineering and Project Execution stage.

- identify risk scenarios associated with low frequency and response of machines in the event of loss of unit or disturbance in the electrical generation system of the Company's plants.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: **Nov 24 2021**
COUNTRY: **Peru**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

During normal operation of the Gas Plant, an electrical fault occurred in the transformer that feeds the Motor Control Centre, producing the sudden shutdown of several main equipment and subsequent total shutdown of the plant (Shut Down Level 3).

WHAT WENT WRONG:

- Inadequate follow-up/closure of control actions: Tasks related to control procedures, not properly implemented.
- Inadequate inspection method/intervals: In the surveys carried out, the issues on which to perform maintenance were not identified.
- Inadequate change management: Documentation used in the last Project was not updated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Properly comply with the Equipment Maintenance Plan.
- Correctly execute and in a timely manner all the actions derived from a MOC.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA ONSHORE

DATE: **Sep 28 2021**
COUNTRY: **Peru**
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Transport - Air
RULE: Line of fire

NARRATIVE:

During boarding of BELL 412 Helicopter, two passengers proceed to approach the Cargo Master who was on the other side in the cargo compartment, passing under the tail of the helicopter. The Cargo Master immediately pulled them away and took them to a safe area. No personal or material damage was reported.

WHAT WENT WRONG:

- Inadequate Initial Training: Personnel do not receive initial induction of air transport operations upon entering the area.
- Confusing/contradictory job/role assignment: EHS coordinator functions are not fully developed due to unification of positions.
- Inadequate management of change process: Management of Change due to the unification of functions/positions of the flight Coordinator and QHSE Coordinator.
- Inadequate identification/evaluation of risks in the development of the standard: Passenger boarding protocol only provides general information for the Cargo Master on the passenger and cargo boarding procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Passengers must:

- Pay attention to the safety videos shown prior to boarding the aircraft.
- Approach the aircraft for boarding only if confirmed by the air operator's personnel.
- It is forbidden to transit by the tail boom of the helicopters - life-threatening risk.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Oct 12 2021
COUNTRY: Peru
FUNCTION: Unspecified
CAUSE: Cut, puncture, scrape
ACTIVITY: Unspecified - other
RULE: Other issue

NARRATIVE:

In the production service workshop, a contractor technician was preparing to manufacture a wooden box measuring 1 X 0.6 X 0.6 metres. He placed the plywood sheet on the work table and, with the help of a colleague who was holding the plywood, cut it using a circular saw. Once said cut was finished, he placed the saw on the table with the blade still turning, and put it away. He did not activate the automatic closing of the blade, which caused the equipment to move towards his left hand, which was about 50cm away and at an approximate angle of 90 degrees, cutting his glove and generating a moderately deep cut wound on his base of the thumb. The technician was transferred to the camp site where first aid was applied.

WHAT WENT WRONG:

- Root cause(s) - Missing or failed safeguards/barriers - Missing or failed Tool/machine safeguards.
- Root cause(s) - Lack of competence assurance.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review and ensure compliance with the supervisory position profile as established in the contract through an audit of the contract.
- Modification and dissemination of the format of the inspection check list of the contractor's tools and equipment that help decision-making and its non-use in case of showing observations.
- Conduct an on-site inspection of all Contractor materials and equipment to identify tools and equipment in poor condition.
- Carry out hazard identification and risk assessment workshops to help strengthen staff and Contractors in the perception of risk related to the activity and the place where it is carried out.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: Dec 8 2021
COUNTRY: Brazil
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Unspecified

NARRATIVE:

A fire occurred in the centrifuge of the cuttings dryer system located in the shaker room, after leakage and ignition of the hydraulic oil of the coupling, during drilling of phase 8 1/2". The fire was put out by rigs with a portable extinguisher and later the deluge system in the shaker room was preventively activated.

WHAT WENT WRONG:

1. Operating system failure not standards, standards and good engineering practices related to Security.
2. Design Failure - Due to centrifuge cylinder installation and locking/Absence of protection device against external closing/Absence of fuse device in the hydraulic coupling.
3. Failure to manage processes - Quality assurance procedure and execution of complicated procedures.
4. Engineering Specification - Hydraulic oil ignition of integrated coupling with flash point in hazardous area.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Issue Safety Alert.
2. Check with the manufacturer for the possibility of reducing the current range of the electric motor relay, in order to cover all risks.
3. Review the Centrifuge Inspection checklist, including operator rounds for checking water supply during centrifuge operation and adding Safety Lockout for valves that may be inadvertently closed.
4. Replacement of hydraulic coupling oil with fire resistant oil.
5. Request to the centrifuge manufacturer to update the manual regarding the use of a hazardous area oil and the addition of the fuse device.
6. Ensure that all equipment manuals for the integrated services are updated in Engineering and in the units.
7. Carry out a study to implement a visit window just below the equipment and in the lines that give access to the sea for inspection of clogging.
8. Carry out an engineering study together with the manufacturer to analyze the feasibility of implementing protection against clogging external to the cylinder.
9. Carry out a study to implement a pressure gauge at the water supply point to facilitate quick visualization by operators.
10. Carry out a study to reposition the emergency button in a place with better access.
11. Establish the asset management control process related to integrated services.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: **Jun 17 2021**
COUNTRY: **Brazil**
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Maintenance, inspection, testing
RULE: Line of fire

NARRATIVE:

During the packer sealing test for acidification of the SGD2 zone (480.5 - 490.5), upon reaching the pressure of 405 psi, there was a rupture of the screws of the arterial ball valve of the production head, and its elements were projected a distance of approximately 21 metres.

WHAT WENT WRONG:

1. Equipment or component failure: corrosion on bolts and valve flange face. There are strong indications that in this specific model of ball valve there is a process of crevice corrosion, favouring premature failure of the screws.
2. Inadequate Maintenance Plan: it was identified that the inspection routine used was not effective to identify the corrosion process in the valve screws and the loss of integrity.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Action to be taken:

1. Include, in the records of surface equipment in interventions, the complete record (manufacturer, model, diameter, others) of all equipment, including side items of the Production Head, as they are critical equipment.
2. Elaborate technical specification to ensure that future ball valve purchases are of the model with screws less susceptible to corrosion by the external environment.
3. Publicize Alert with recommendation to change ball valves of the specific model, based on informative documentation.
4. Develop an informative manual for the operation of surface equipment, covering operational and preventive care, aiming at maintaining the integrity of the equipment.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: May 30 2021
COUNTRY: Brazil
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

At 01.00 a back-loading operation with a boat had begun with the East Crane. The crane operator moved the crane to the first load. While swinging to load to the boat the crane operator felt vibrations in his controls. He immediately stopped the crane from swinging and heard a loud noise. Deck crew noted that the maintenance jib had detached from the top of the crane and fell 30/35 metres (193kg) in an area opposite the crane boom on the drilling deck. After the crane was inspected, the crane operator landed the cargo back on the Pipe deck and lifted the crane in the same position for additional inspection.

No injuries and no one was present in the area.

WHAT WENT WRONG:

Underlying Causes

Failure in technical barrier – Offshore crane – other barriers for deck cranes:

- Not setting the locking pin mechanism in place after use and lack of procedure to use it. On the other hand, despite having been evaluated with low probability, the locking pin could have been detached by strong winds.
- Lack of specific verification of jib crane in periodic (pre-operation, daily, weekly, monthly) crane routines.
- Inadequate inspection and maintenance programs:
- Verification of jib crane was not present in periodic crane routines.
- Jib crane remained out of position for more than a month and it was not observed during observation/safety rounds.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Include a routine in monthly maintenance work orders to check locking pins for both cranes.
- Include verification of Jib crane on pedestal crane periodic inspection checklist.
- Include verification of Jib crane on pedestal crane pre-operation checklists and DROPS book/checklist.
- Establish specific checklist to be attached to the Permit to Work Mechanical handling plan for maintenance when it is defined to use jib cranes for maintenance/services.
- Installation of a counter-pin to L-shaped locking pin.
- Consider refresh Inspection and Observation techniques training or similar for crane operation and maintenance crews.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: **Feb 7 2021**
COUNTRY: **Brazil**
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Other issue

NARRATIVE:

At around 08:30 the scaffolding team went to the area to open the WP. Upon arrival, it was observed a dropped handrail section on the first level of the module 30, partially over an escape route, which had fallen from the second mezzanine level (davit platform). The nearby area on main deck had normally been barricaded-off due to diving operations, but not the mezzanine where the handrail section fell. It is believed that the handrail had fallen overnight when no activities were being held on the location. There were no eyewitness from the exact moment of the fall and no CCTV footage was available. The handrail weighted around 49 kg and fell 5 metres, resulting in an energy release of 2403 J.

WHAT WENT WRONG:

- Technical failures due to ageing: with the known environmental conditions present, aligned with the absence of any conservation treatment over the years, the natural outcome ended-up being the ageing of the material through corrosion.
- Maintenance has not been performed in accordance with programme and known defects not corrected.
- On maintenance, planning, and operations teams, there had been the perception that maintaining these structures was not safety critical, which might indeed have been true for many years. However, the consequences of cancelling and systematically postponing these conservation activities were inadequately assessed and contributed to a severe degradation of the handrail's condition throughout the years.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Assess the improvement of technical specifications of handrails and structures to avoid dissimilar materials minimizing the risk of galvanic corrosion.
- Provide orientation to the maintenance planning team on how to effectively raise and manage notifications and WO to improve traceability of corrosion. These shall not be closed until the conclusive execution of the work, nor replaced with new ones.
- Assess implementing a permanent conservation crew onboard to attack immediate risk issues that are found and reported.
- Assess having a permanent supply of conservation material available onboard for immediate conservation/ replacement work.
- Execute the corrosion management strategy for surface mechanical treatment under regular periods (SOLV campaigns).
- Evaluate new ways to monitor and risk assess non-critical overdue WO related to the System 2 maintenance to support management decision on the prioritization of such work, in a way to better mirror installation actual conditions.
- Reinforce with the work force the importance of proactively reporting unsafe conditions and to address the right classifications on the right levels, with SSU involvement, to ensure the incidents are described and classified adequately and investigated into the appropriate levels.

SOUTH & CENTRAL AMERICA OFFSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: Feb 20 2021
COUNTRY: Brazil
FUNCTION: Production
CAUSE: Explosion
ACTIVITY: Transport - Water, incl. marine activity
RULE: Energy isolation

NARRATIVE:

A nitrogen cylinder, from a Production Platform, inside a specific package with 8 others cylinders, exploded on the deck of a Platform Supply Vessel (PSV), nearly hitting 2 crew members on the deck.

WHAT WENT WRONG:

Cylinders inspection expired.

The cylinder was stored in a inadequate area and didn't meet the Material Safety Data Sheet (MSDS) recommendations.

Inadequate mooring caused mechanical shock between the cylinders or between the cylinders and its package.

The rise of temperature and consequent rise in internal gas pressure of the cylinder.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Pressurized cylinders must be submitted to a periodic inspection program, in order to meet the maximum intervals between inspections in accordance with NBR-12274.

Pressurized cylinders must meet recommendations of their Material Safety Data Sheet (MSDS).

Attention to storage guidelines, especially in relation to exposure to weather and high temperature.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: Jul 3 2021
COUNTRY: Brazil
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Lifting, crane, rigging, deck operations
RULE: Safe mechanical lifting

NARRATIVE:

During an operation with the crane, the pin of the first cable guide roller of the auxiliary winch came off, causing the 1.8 kg roller to fall from a height of 23 metres, onto the main deck.

WHAT WENT WRONG:

1. Assembly failure (absence or improper assembly of lock washers).
2. Non-compliance with the Maintenance Plan (no use on board).
3. Inadequate Maintenance Plan (to-do list detail).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Action to be taken:

1. Check and correctly assemble the pair of lock washers on all stern and bow crane roller axles.
2. Mobilize fourth component of the IRATA team with mechanical training.
3. Correct the list of tasks in the plans, making it clear that the correct assembly of the pair of rim lock washers must be checked.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: Sep 30 2021
COUNTRY: Trinidad & Tobago
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

A scaffold crew of 3 were tasked with removing scaffolding from the well production platform (WPP). In order to remove scaffolding the hatch cover had to be reinstalled first. The process was to install the hatch cover in two segments. Both hatch cover segments were stored adjacent to the slot and were slid over by attaching independent rope to each of the four eye bolts with 3 scaffold builders manoeuvring them into place and resting on the support beams. The initial positioning of the 1st hatch segment was off centre from the support beam. The crew then attempted to reposition the hatch segment by pulling one of the eyebolts by its rope while the other 2 supported as it rested on the frame. At this point the eyebolt that was being pulled on failed, resulting in the hatch cover falling 8' to the deck below which was an empty slot with no well head below.

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WHAT WENT WRONG:

The eyebolt that was being pulled on failed, resulting in the hatch cover falling 8' to the deck below.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Remove and replace lifting eye bolts on the WPP hatch covers according to manufacture specs for rated load capacities and angle.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: Oct 22 2021
COUNTRY: Trinidad & Tobago
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Other issue

NARRATIVE:

After days of adverse weather, a team member was walking on the upper mast deck when he noticed a section from the back scratcher and a peg laying on the deck under the mast. At the time, no one was in the area as it is not highly traversed. One of the items weighed 6 pounds and fell 15 metres while the other weighed 2 pounds and fell 14.5 metres; both outcomes on the DROPS calculator indicated a potential fatality in the event the objects struck someone.

WHAT WENT WRONG:

A section from the back scratcher and a peg were dislodged during weather event.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

A verification of the management of deficiencies from the 3rd Party DROPS Inspection Report may have identified that the corrective maintenance work orders generated to address the deficiencies were not being actioned.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: Nov 15 2021
COUNTRY: Trinidad & Tobago
FUNCTION: Drilling
CAUSE: Unspecified - Other
ACTIVITY: Drilling, workover, well services
RULE: Line of fire

NARRATIVE:

When running in the hole with the landing string on elevators, a leak was noticed while making up to the drill string. While breaking out the drill string for racking back the stand to investigate the leak, the driller noticed something pass the camera. Immediately afterwards a mud flex hose contacted the handrail on the starboard side of the ZGA track. The rig floor red zone was clear, and all personnel were in the safe step back area, per job design.

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WHAT WENT WRONG:

Mud hose broke loose while breaking out the drill string.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

No requirements for potential dropped object checks prior to vendor equipment being transported offshore for use.

Update switch assembly procedure to require fittings and secondary retention.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: May 22 2021
COUNTRY: Trinidad & Tobago
FUNCTION: Production
CAUSE: Unspecified - Other
ACTIVITY: Production operations
RULE: Other issue

NARRATIVE:

A total of twenty seven (27) persons tested positive for COVID-19 who were at offshore facility.

WHAT WENT WRONG:

COVID-19 Field Operations Emergency Response plan developed/revised without review/approval.

This was supported by the identification of several gaps: lack of consultation with SMEs, inaccurate incorporation of guidance information referenced or placed into documents, and the absence of documentation of critical work activities performed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Test market for and evaluate more reliable COVID tests.
- Conduct refresher training course for all operational sites personnel on reporting any illness, injuries and allergies to Medic upon arrival to operational sites.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

International Association of Oil & Gas Producers

For the full analysis of 2021 results, as well as all fatal incident reports submitted to IOGP since 1991, visit IOGP's data website at: https://data.iogp.org

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